# Introduction to SQL for BigQuery and Cloud SQL - LAB

**GSP281**

Google Cloud Self-Paced Labs

**Overview**

SQL (Structured Query Language) is a standard language for data operations that allows you to ask questions and get insights from structured datasets. It's commonly used in database management and allows you to perform tasks like transaction record writing into relational databases and petabyte-scale data analysis.

This lab is divided into two parts: in the first half, you will learn fundamental SQL querying keywords, which you will run in BigQuery on a public dataset that contains information on London bikeshares.

In the second half, you will learn how to export subsets of the London bikeshare dataset into CSV files, which you will then upload to Cloud SQL. From there you will learn how to use Cloud SQL to create and manage databases and tables. Towards the end, you will get hands-on practice with additional SQL keywords that manipulate and edit data.

## What you'll learn

In this lab, you will learn how to:

- Read data into BigQuery.
- Execute simple queries in BigQuery to explore data.
- Export a subset of data into a CSV file and store that file in a new Cloud Storage bucket.
- Create a new Cloud SQL instance and load your exported CSV file as a new table.

**Caution:** When you are in the console, do not deviate from the lab instructions. Doing so may cause your account to be blocked. Learn more.

Open Google Cloud console

**Username**

student-01-a74bdd6d45a3(

**Password**

YS1vmA0FWOE3

**Project ID**

qwiklabs-gcp-00-cd593d3:

# Task 1. The basics of SQL

## Databases and tables

As mentioned earlier, SQL allows you to get information from "structured datasets". Structured datasets have clear rules and formatting and often times are organized into tables, or data that's formatted in rows and columns.

An example of *unstructured data* would be an image file. Unstructured data is inoperable with SQL and cannot be stored in BigQuery datasets or tables (at least natively.) To work with image data (for instance), you would use a service like Cloud Vision, perhaps through its API directly.

The following is an example of a structured dataset—a simple table:

| User | Price | Shipped |
|---|---|---|
| Sean | $35 | Yes |

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# Task 2. Exploring the BigQuery console

## The BigQuery paradigm

BigQuery is a fully-managed petabyte-scale data warehouse that runs on the Google Cloud. Data analysts and data scientists can quickly query and filter large datasets, aggregate results, and perform complex operations without having to worry about setting up and managing servers. It comes in the form of a command line tool (pre installed in cloudshell) or a web console—both ready for managing and querying data housed in Google Cloud projects.

In this lab, you use the web console to run SQL queries.

## Open the BigQuery console

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housed in Google Cloud projects.

In this lab, you use the web console to run SQL queries.

## Open the BigQuery console

1. In the Google Cloud Console, select **Navigation menu** > **BigQuery**.

The **Welcome to BigQuery in the Cloud Console** message box opens. This message box provides a link to the quickstart guide and the release notes.

2. Click **Done**.

The BigQuery console opens.

Take a moment to note some important features of the UI. The right-hand side of the console houses the query "Editor". This is where you write and run SQL commands like the examples covered earlier. Below that is "Query history", which is a list of queries you ran previously.

The left pane of the console is the **Navigation menu**. Apart from the self-explanatory

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BigQuery – qwiklabs-gcp-00-cd5

https://console.cloud.google.com/bigquery?referrer=search&project=qwiklabs-gcp-00-cd593d3377bb&ws=!1m0

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Show starred only

qwiklabs-gcp-00-cd593d33...

Untitled...ery

## Create new

What's new in Studio

SQL query | Notebook | Notebook with Spark | ML model | Data canvas | Pipeline | Data preparation | Table

## Recent

| Display name | Type | Last modified time | Project |
| --- | --- | --- | --- |
| No rows to display | | | |

## Try with templates

View notebook gallery →

### Try the Google Trends Demo Query

This simple query generates the top search terms in the US from the Google Trends public dataset.

Open View dataset

### Try the Colab Demo Notebook

This notebook walks you through their basics and showcases BigQuery DataFrames.

Open

## Job history

Show

Trending videos Stranger Things... Search ENG IN 04:58 PM 27-11-2025

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End Lab 01:09:59

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Open Google Cloud console

**Username**

student-01-a74bdd6d45a3(

**Password**

YS1vmA0FWOE3

**Project ID**

qwiklabs-gcp-00-cd593d3:

intricacies of the console, you can load up some queryable data.

## Uploading queryable data

In this section you pull in some public data into your project so you can practice running SQL commands in BigQuery.

1. Navigate back to **Explorer** and click **Add data**.
2. Choose **Star a project by name**.
3. Enter project name as **bigquery-public-data**.
4. Click **STAR**.

It's important to note that you are still working out of your lab project in this new tab. All you did was pull a publicly accessible project that contains datasets and tables into BigQuery for analysis — you didn't *switch over* to that project. All of your jobs and services are still tied to your Google Cloud Skills Boost account. You can see this for yourself by inspecting the project field near the top of the console:

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Try the Spark Demo Notebook

This notebook walks you through their basics and showcases Spark in BigQuery.

Open

## Add your own data

**Local file**

Upload a local file

Launch this guide

**Google Drive**

Google storage service

Launch this guide

**Google Cloud Storage**

Google object storage service

Launch this guide

Show home page on startup

Job history | Show

20°C Sunny | Search | ENG IN | 05:00 PM 27-11-2025

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**Google Drive**
Google storage service
Launch this guide

**Google Cloud Storage**
Google object storage service
Launch this guide

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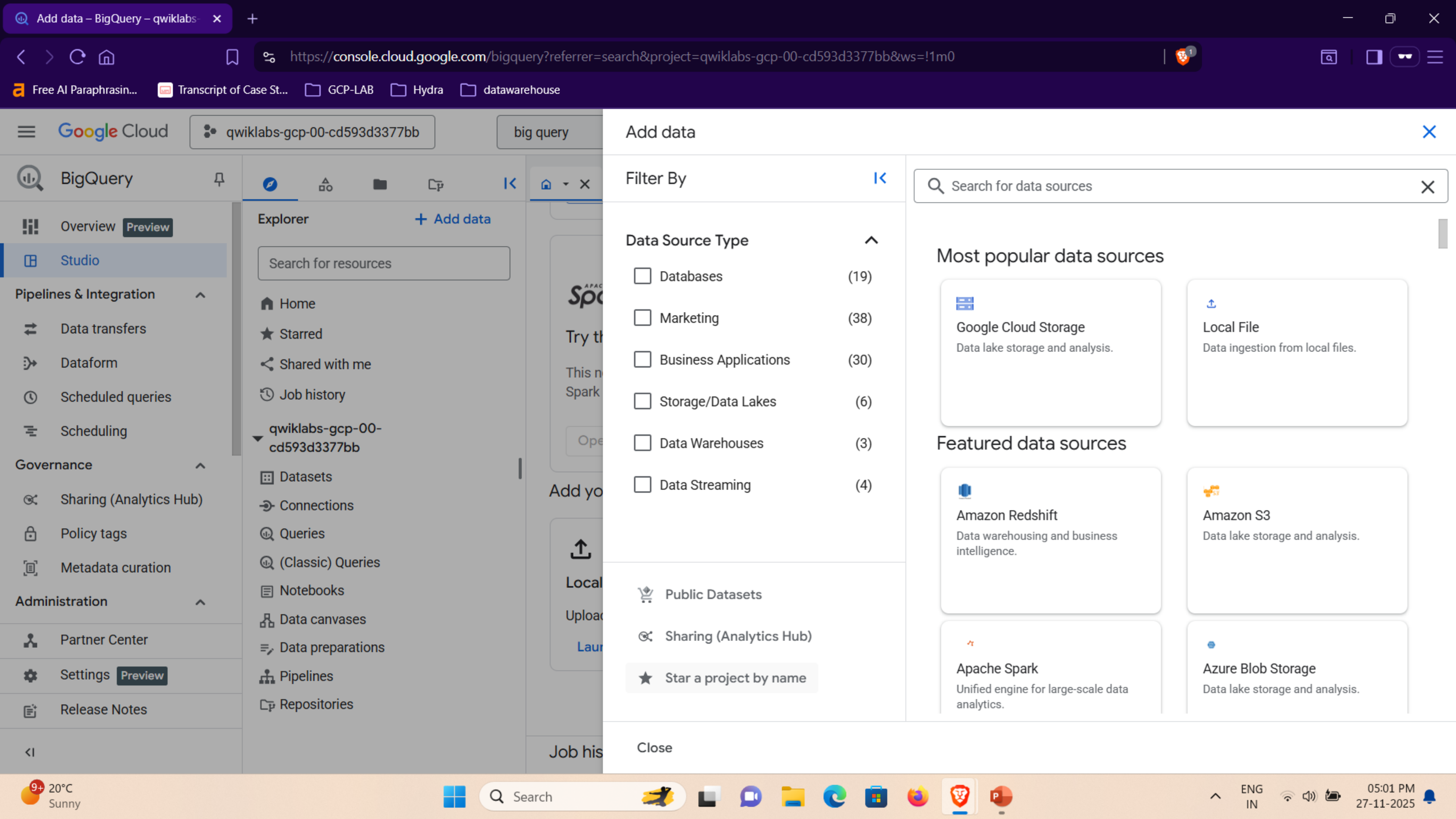
Add data – BigQuery – qwiklabs-
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Try the Spark Demo Notebook

This no... ...d showcases Spark i...

Oper...

## Star a project

Project name *

bigquery-public-data

Cancel | Star

Add you...

**Local file**

Upload a local file

Launch this guide

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Google storage service

Launch this guide

**Google Cloud Storage**

Google object storage service

Launch this guide

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Lab setup instructions and requirements

End Lab 01:04:37

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Open Google Cloud console

Username
student-01-a74bdd6d45a3(

Password
YS1vmA0FWOE3

Project ID
qwiklabs-gcp-00-cd593d3:

BI Engine
▸ bigquery-public-data

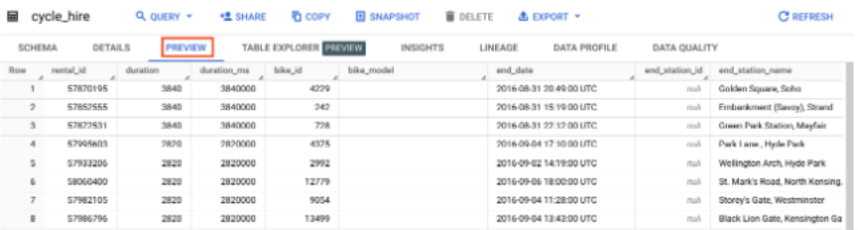
5. You now have access to the following data:

- Google Cloud Project → `bigquery-public-data`
- Dataset → `london_bicycles`

6. Click on the **london bicycles** dataset to reveal the associated tables

- Table → `cycle_hire`
- Table → `cycle_stations`

In this lab you will use the data from **cycle_hire**. Open the `cycle_hire` table, then click the **Preview** tab. Your page should resemble the following:



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Datasets – BigQuery – qwiklabs-

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| | | | |
|---|---|---|---|
| iowa_liquor_sales_forecasting | Default | US | dg_data_source: public |
| irs_990 | Default | US | dg_data_source: public |
| labeled_patents | Default | US | dg_data_confidentiality: public |
| libraries_io | Default | US | dg_data_source: public |
| listenbrainz | Default | US | dg_data_source: public |
| london_bicycles | Default | EU | dg_data_source: public |
| | Default | EU | dg_data_source: public |
| | Default | EU | dg_data_source: public |
| | Default | US | dg_data_source: public |
| medicare | Default | US | dg_data_source: public |
| ml_datasets | Default | US | dg_data_confidentiality: public |
| ml_datasets_uscentral1 | Default | us-central1 | dg_data_source: public |
| modis_terra_net_primary_production | Default | US | dg_data_source: public |
| moon_phases | Default | US | dg_data_source: public |
| multilingual_spoken_words_corpus | Default | US | dg_data_source: public |
| nasa_wildfire | Default | US | dg_data_source: public |
| national_water_model | Default | US | dg_data_confidentiality: public |
| ncaa_basketball | Default | US | dg_data_source: public |
| nces_ipeds | Default | US | dg_data_source: public |
| new_york | Default | US | dg_data_source: public |
| new_york_311 | Default | US | dg_data_source: public |
| new_york_citibike | Default | US | dg_data_source: public |
| new_york_mv_collisions | Default | US | dg_data_source: public |

london_bicycles
Click to open in current tab
(Ctrl + Click - open in new tab,
Shift + Ctrl + Click - open in split tab)

Rows per page: 50 | 201 – 250 of many

Job history | Show

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london_bicycles – BigQuery – qw

https://console.cloud.google.com/bigquery?referrer=search&project=qwiklabs-gcp-00-cd593d3377bb&ws=!1m4!1m3!3m2!1sbigquery-public-d...

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Untitled...ery | london_...les

bigquery-public-data / Datasets / london_bicycles

## london_bicycles

Create Table | Share | Copy | Delete | Refresh

Overview | Details

Tables | Models | Routines

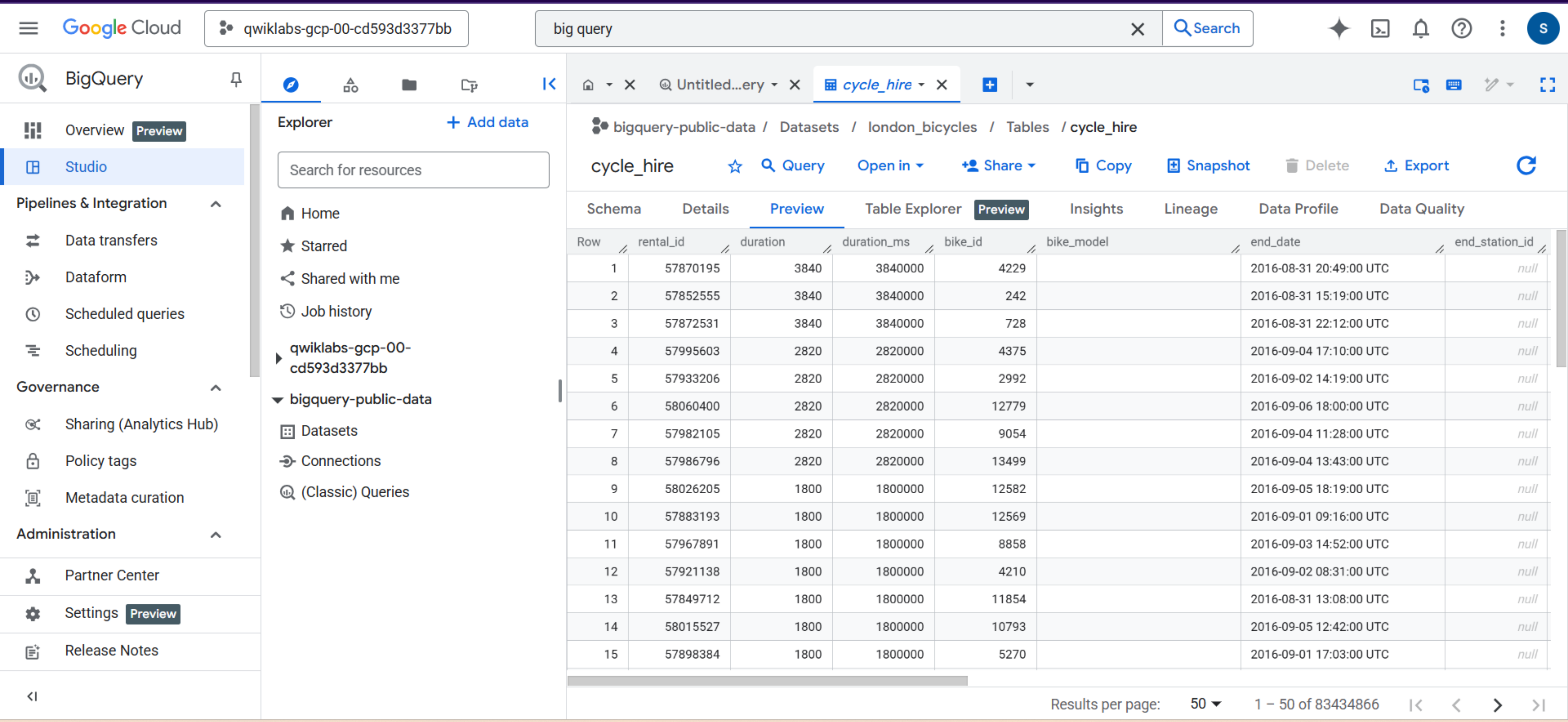
Filter Enter property name or value

| Table ID | Type | Create time | Expiration time |
|---|---|---|---|
| cycle_hire | Table | May 3, 20... | None |
| cycle_stations | Table | Jul 27, 20... | None |

Rows per page: 50 1 – 2 of 2

Job history Show

20°C Sunny | Search | ENG IN | 05:04 PM 27-11-2025




| Row | rental_id | duration | duration_ms | bike_id | bike_model | end_date | end_station_id |
|---|---|---|---|---|---|---|---|
| 1 | 57870195 | 3840 | 3840000 | 4229 | | 2016-08-31 20:49:00 UTC | null |
| 2 | 57852555 | 3840 | 3840000 | 242 | | 2016-08-31 15:19:00 UTC | null |
| 3 | 57872531 | 3840 | 3840000 | 728 | | 2016-08-31 22:12:00 UTC | null |
| 4 | 57995603 | 2820 | 2820000 | 4375 | | 2016-09-04 17:10:00 UTC | null |
| 5 | 57933206 | 2820 | 2820000 | 2992 | | 2016-09-02 14:19:00 UTC | null |
| 6 | 58060400 | 2820 | 2820000 | 12779 | | 2016-09-06 18:00:00 UTC | null |
| 7 | 57982105 | 2820 | 2820000 | 9054 | | 2016-09-04 11:28:00 UTC | null |
| 8 | 57986796 | 2820 | 2820000 | 13499 | | 2016-09-04 13:43:00 UTC | null |
| 9 | 58026205 | 1800 | 1800000 | 12582 | | 2016-09-05 18:19:00 UTC | null |
| 10 | 57883193 | 1800 | 1800000 | 12569 | | 2016-09-01 09:16:00 UTC | null |
| 11 | 57967891 | 1800 | 1800000 | 8858 | | 2016-09-03 14:52:00 UTC | null |
| 12 | 57921138 | 1800 | 1800000 | 4210 | | 2016-09-02 08:31:00 UTC | null |
| 13 | 57849712 | 1800 | 1800000 | 11854 | | 2016-08-31 13:08:00 UTC | null |
| 14 | 58015527 | 1800 | 1800000 | 10793 | | 2016-09-05 12:42:00 UTC | null |
| 15 | 57898384 | 1800 | 1800000 | 5270 | | 2016-09-01 17:03:00 UTC | null |

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End Lab 01:03:13

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student-01-a74bdd6d45a3

**Password**

YS1vmA0FWOE3

**Project ID**

qwiklabs-gcp-00-cd593d3

## Running SELECT, FROM, and WHERE in BigQuery

You now have a basic understanding of SQL querying keywords and the BigQuery data paradigm and some data to work with. Run some SQL commands using this service.

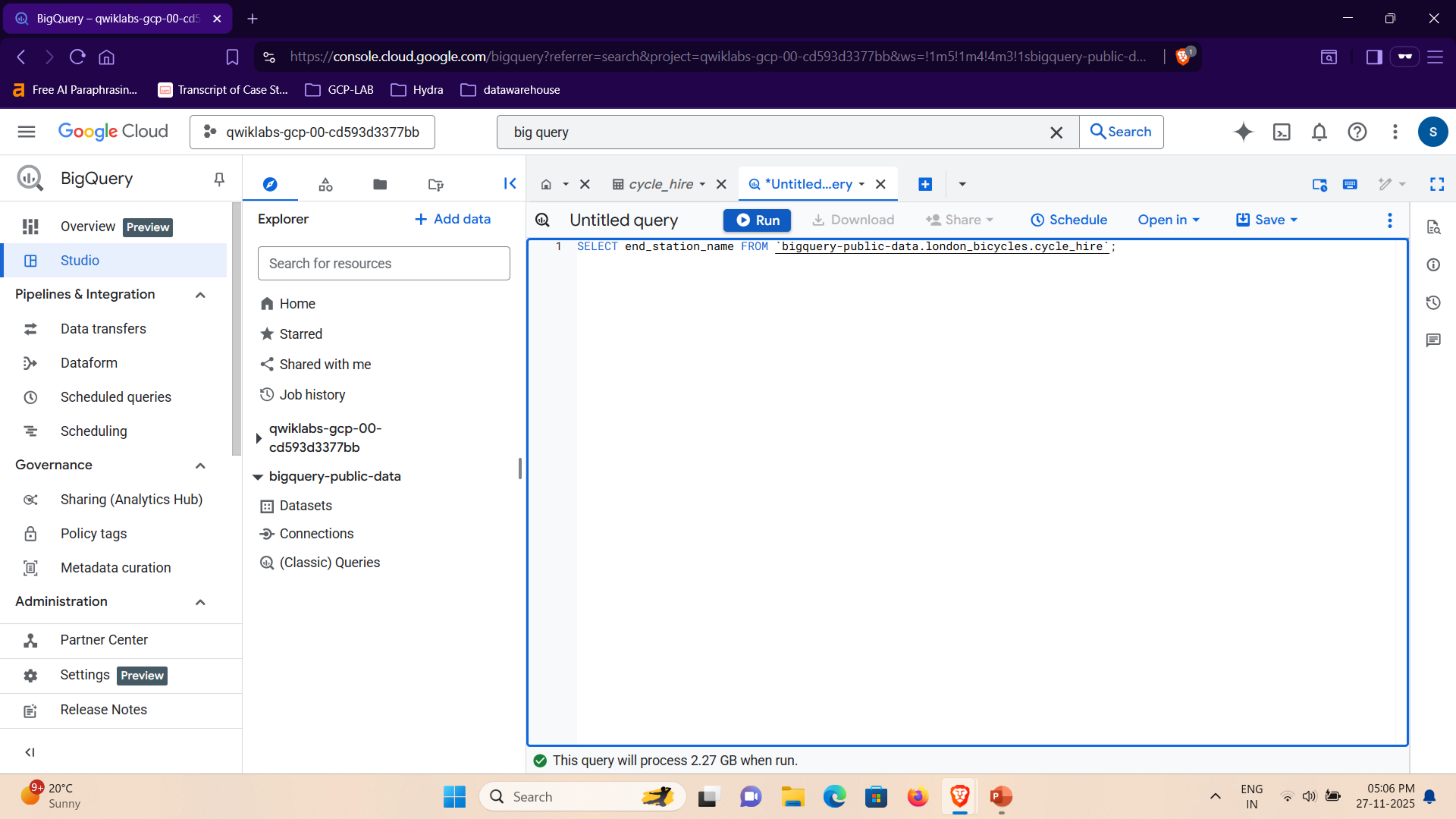
If you look at the bottom right corner of the console, you will notice that there are **83,434,866** rows of data, or individual bikeshare trips taken in London between 2015 and 2017 (not a small amount by any means!)

Now take note of the ninth column key: `end_station_name`, which specifies the end destination of bikeshare rides. Before getting too deep, run a simple query to isolate the `end_station_name` column.

1. Copy and paste the following command into the query **Editor**:

```
SELECT end_station_name FROM `bigquery-public-
data.london_bicycles.cycle_hire`;
```

2. Then click **Run**.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*Untitled...ery
Untitled query
Run
Download
Share
Schedule
Open in
Save
1 SELECT end_station_name FROM `bigquery-public-data.london_bicycles.cycle_hire`;
This query will process 2.27 GB when run.
20°C
Sunny
Search
ENG
IN
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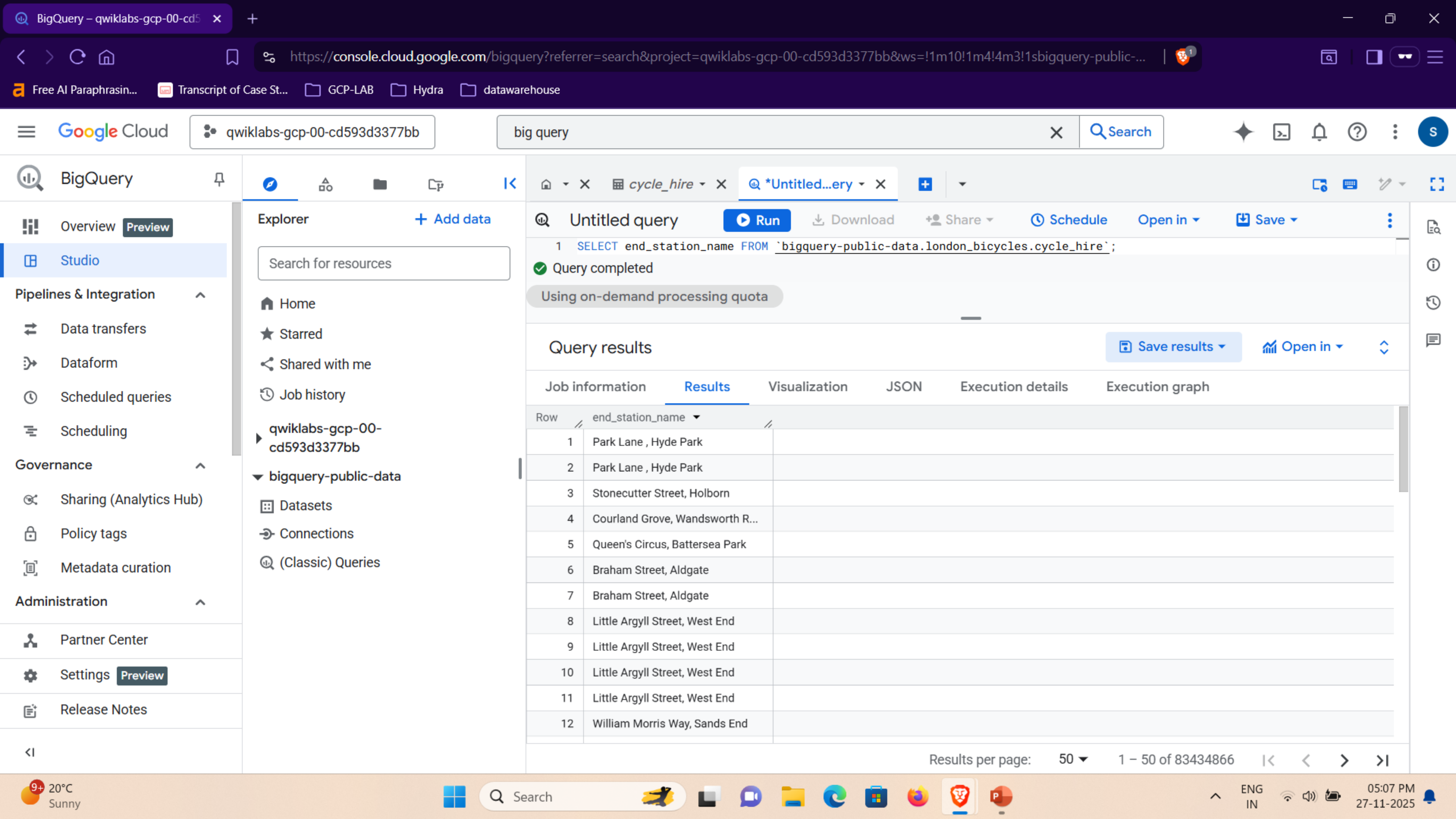
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1 SELECT end_station_name FROM `bigquery-public-data.london_bicycles.cycle_hire`;
Query completed
Using on-demand processing quota
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Row
end_station_name
1 Park Lane , Hyde Park
2 Park Lane , Hyde Park
3 Stonecutter Street, Holborn
4 Courland Grove, Wandsworth R...
5 Queen's Circus, Battersea Park
6 Braham Street, Aldgate
7 Braham Street, Aldgate
8 Little Argyll Street, West End
9 Little Argyll Street, West End
10 Little Argyll Street, West End
11 Little Argyll Street, West End
12 William Morris Way, Sands End
Results per page: 50
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End Lab 01:00:10

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Open Google Cloud console

Username
student-01-a74bdd6d45a3(

Password
YS1vmA0FWOE3

Project ID
qwiklabs-gcp-00-cd593d3:

Why don't you find out how many bike trips were 20 minutes or longer?

3. Clear the query from the editor, then run the following query that utilizes the `WHERE` keyword:

```
SELECT * FROM `bigquery-public-data.london_bicycles.cycle_hire`
WHERE duration>=1200;
```

This query may take a minute or so to run.

`SELECT *` returns all column values from the table. Duration is measured in seconds, which is why you used the value 1200 (60 * 20).

If you look in the bottom right corner you see that **26,441,016** rows were returned. As a fraction of the total (26441016/83434866), this means that ~30% of London bikeshare rides lasted 20 minutes or longer (they're in it for the long haul!)

## Test your understanding

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cycle_hire | *Untitled...ery

Untitled query | Run | Download | Share | Schedule | Open in | Save

```
SELECT * FROM `bigquery-public-data.london_bicycles.cycle_hire` WHERE duration>=1200;
```

Query completed

Using on-demand processing quota

## Query results

Save results | Open in

Job information | Results | Visualization | JSON | Execution details | Execution graph

| Row | rental_id | duration | duration_ms | bike_id | bike_model | end_date |
|---|---|---|---|---|---|---|
| 1 | 125128827 | 1518 | 1517816 | 40680 | CLASSIC | 2022-09-15 08:56:00 UTC |
| 2 | 68808746 | 9898 | 9898000 | 4741 | null | 2017-08-26 16:56:00 UTC |
| 3 | 42538208 | 3660 | 3660000 | 4159 | | 2015-04-12 17:55:00 UTC |
| 4 | 126687079 | 29453 | 29453013 | 22437 | CLASSIC | 2022-11-10 21:49:00 UTC |
| 5 | 101392998 | 4560 | 4560000 | 11304 | null | 2020-09-02 19:32:00 UTC |
| 6 | 48341685 | 6480 | 6480000 | 9920 | | 2015-10-01 09:01:00 UTC |
| 7 | 52494865 | 8460 | 8460000 | 9285 | | 2016-03-30 17:14:00 UTC |
| 8 | 123961548 | 3480 | 3480000 | 20535 | null | 2022-08-12 22:32:00 UTC |
| 9 | 120896306 | 3900 | 3900000 | 349 | null | 2022-06-03 11:59:00 UTC |
| 10 | 42616102 | 51900 | 51900000 | 5149 | | 2015-04-15 22:22:00 UTC |
| 11 | 42627155 | 4620 | 4620000 | 12725 | | 2015-04-15 14:20:00 UTC |
| 12 | 120487393 | 3360 | 3360000 | 6685 | null | 2022-05-23 17:21:00 UTC |

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End Lab 00:59:19

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Open Google Cloud console

Username

student-01-a74bdd6d45a3(

Password

YS1vmA0FWOE3

Project ID

qwiklabs-gcp-00-cd593d3:

# Task 3. More SQL Keywords: GROUP BY, COUNT, AS, and ORDER BY

## GROUP BY

The `GROUP BY` keyword will aggregate result-set rows that share common criteria (e.g. a column value) and will return all of the unique entries found for such criteria.

This is a useful keyword for figuring out categorical information on tables.

1. To get a better picture of what this keyword does, clear the query from the editor, then copy and paste the following command:

```
SELECT start_station_name FROM `bigquery-public-
data.london_bicycles.cycle_hire` GROUP BY start_station_name;
```

0/100

Lab setup instructions and requirements

End Lab 00:58:44

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Open Google Cloud console

Username

student-01-a74bdd6d45a3(

Password

YS1vmA0FWOE3

Project ID

qwiklabs-gcp-00-cd593d3:

The `GROUP BY` keyword will aggregate result-set rows that share common criteria (e.g. a column value) and will return all of the unique entries found for such criteria.

This is a useful keyword for figuring out categorical information on tables.

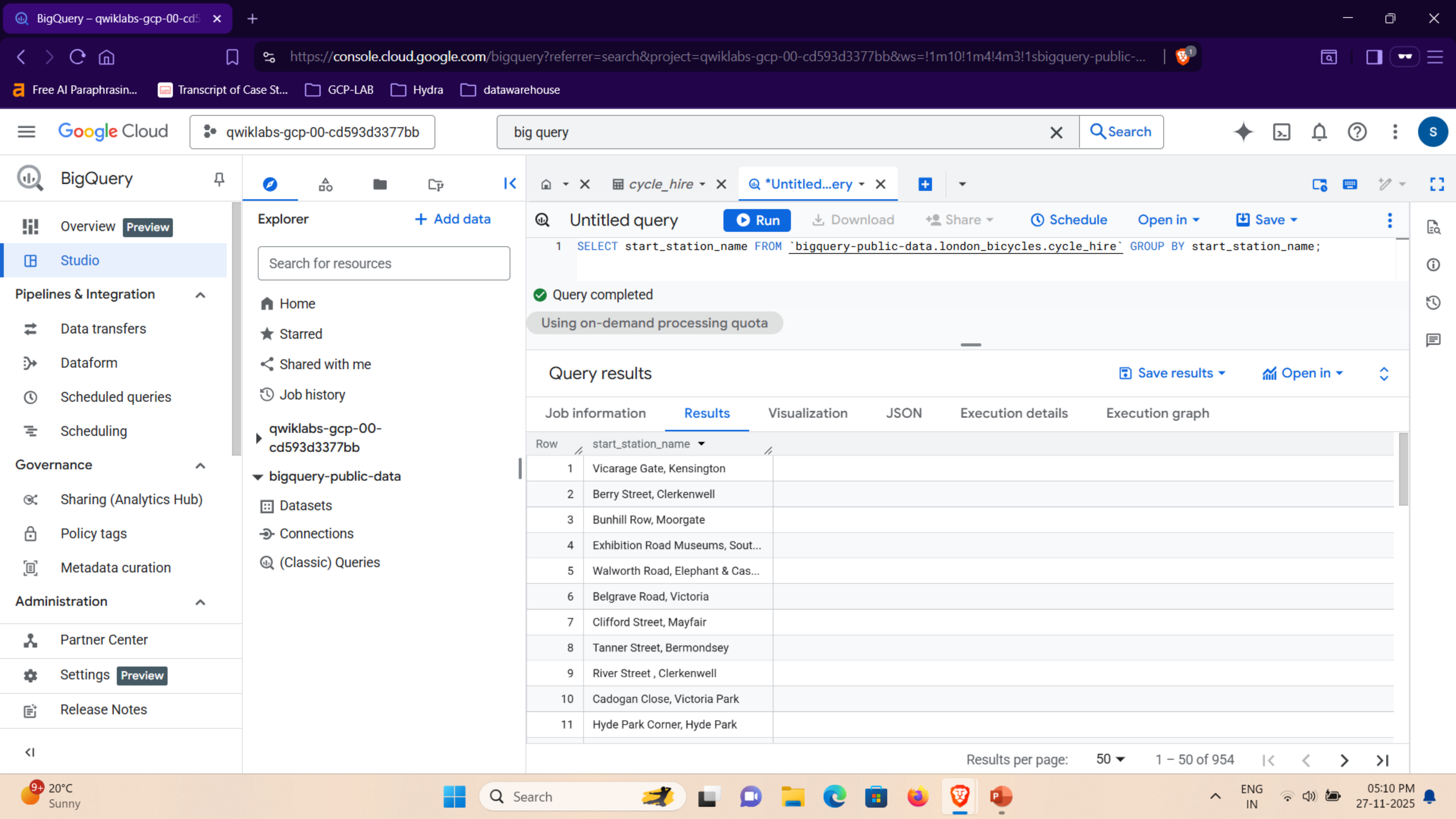
1. To get a better picture of what this keyword does, clear the query from the editor, then copy and paste the following command:

```
SELECT start_station_name FROM `bigquery-public-
data.london_bicycles.cycle_hire` GROUP BY start_station_name;
```

2. Click **Run**.

Your results are a list of unique (non-duplicate) column values.

Without the `GROUP BY`, the query would have returned the full **83,434,866** rows. `GROUP BY` will output the unique column values found in the table. You can see this for yourself by looking in the bottom right corner. You will see **954** rows, meaning there are 954 distinct London bikeshare starting points.



BigQuery – qwiklabs-gcp-00-cd5
https://console.cloud.google.com/bigquery?referrer=search&project=qwiklabs-gcp-00-cd593d3377bb&ws=!1m10!1m4!4m3!1sbigquery-public-...
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1 SELECT start_station_name FROM `bigquery-public-data.london_bicycles.cycle_hire` GROUP BY start_station_name;
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1 Vicarage Gate, Kensington
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4 Exhibition Road Museums, Sout...
5 Walworth Road, Elephant & Cas...
6 Belgrave Road, Victoria
7 Clifford Street, Mayfair
8 Tanner Street, Bermondsey
9 River Street , Clerkenwell
10 Cadogan Close, Victoria Park
11 Hyde Park Corner, Hyde Park
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Lab setup instructions and requirements

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Open Google Cloud console

**Username**
student-01-a74bdd6d45a3(

**Password**
YS1vmA0FWOE3

**Project ID**
qwiklabs-gcp-00-cd593d3:

distinct London bikeshare starting points.

## COUNT

The `COUNT()` function will return the number of rows that share the same criteria (e.g. column value). This can be very useful in tandem with a `GROUP BY`.

Add the `COUNT` function to our previous query to figure out how many rides begin at each starting point.

- Clear the query from the editor, then copy and paste the following command and then click **Run**:

```
SELECT start_station_name, COUNT(*) FROM `bigquery-public-
data.london_bicycles.cycle_hire` GROUP BY start_station_name;
```

Your output shows how many bikeshare rides begin at each starting location.



0/100

BigQuery – qwiklabs-gcp-00-cd5

https://console.cloud.google.com/bigquery?referrer=search&project=qwiklabs-gcp-00-cd593d3377bb&ws=!1m10!1m4!4m3!1sbigquery-public-...

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cycle_hire | *Untitled...ery

Untitled query | Run | Download | Share | Schedule | Open in | More | Save

```
SELECT start_station_name, COUNT(*) FROM `bigquery-public-data.london_bicycles.cycle_hire` GROUP BY start_station_name;
```

Query completed

Using on-demand processing quota

Query results | Save results | Open in

Job information | Results | Visualization | JSON | Execution details | Execution graph

| Row | start_station_name | f0_ |
|---|---|---|
| 1 | Exhibition Road Museums, Sout... | 102724 |
| 2 | Bow Church Station, Bow | 98575 |
| 3 | Hyde Park Corner, Hyde Park | 671688 |
| 4 | South Audley Street, Mayfair | 45852 |
| 5 | Bayley Street , Bloomsbury | 148392 |
| 6 | Finsbury Circus, Liverpool Street | 263926 |
| 7 | Montgomery Square, Canary W... | 20809 |
| 8 | Southwark Street, Bankside | 56572 |
| 9 | Haggerston Road, Haggerston | 127714 |
| 10 | Albert Bridge Road, Battersea P... | 131622 |
| 11 | Berry Street, Clerkenwell | 172221 |

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20°C Sunny | Search | ENG IN | 05:11 PM 27-11-2025

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End Lab 00:56:41

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Open Google Cloud console

**Username**
student-01-a74bdd6d45a3(

**Password**
YS1vmA0FWOE3

**Project ID**
qwiklabs-gcp-00-cd593d3:

## AS

SQL also has an `AS` keyword, which creates an *alias* of a table or column. An alias is a new name that's given to the returned column or table—whatever `AS` specifies.

1. Add an `AS` keyword to the last query you ran to see this in action. Clear the query from the editor, then copy and paste the following command:

```
SELECT start_station_name, COUNT(*) AS num_starts FROM
`bigquery-public-data.london_bicycles.cycle_hire` GROUP BY
start_station_name;
```

2. Click **Run**.

For Results, the right column name changed from `COUNT(*)` to `num_starts`.

As you see, the `COUNT(*)` column in the returned table is now set to the alias name `num_starts`. This is a handy keyword to use especially if you are dealing with large sets of data — forgetting that an ambiguous table or column name happens more often than you think!

0/100

Untitled query

Run | Download | Share | Schedule | Open in | More | Save

```
SELECT start_station_name, COUNT(*) AS num_starts FROM `bigquery-public-data.london_bicycles.cycle_hire` GROUP BY start_station_name;
```

Query completed

Query results

Save results | Open in

Job information | Results | Visualization | JSON | Execution details | Execution graph

| Row | start_station_name | num_starts |
|---|---|---|
| 1 | Salmon Lane, Limehouse | 87352 |
| 2 | Millennium Hotel, Mayfair | 61166 |
| 3 | Watney Street, Shadwell | 120980 |
| 4 | Harriet Street, Knightsbridge | 115431 |
| 5 | Carnegie Street, King's Cross | 73330 |
| 6 | Rodney Road , Walworth | 105665 |
| 7 | Percival Street, Finsbury | 88647 |
| 8 | Southwark Station 2, Southwark | 122640 |
| 9 | St. James's Square, St. James's | 221807 |
| 10 | Holy Trinity Brompton, Knightsb... | 103862 |
| 11 | Clarkson Street, Bethnal Green | 49805 |

Results per page: 50 1 – 50 of 954

## ORDER BY

The `ORDER BY` keyword sorts the returned data from a query in ascending or descending order based on a specified criteria or column value. Add this keyword to our previous query to do the following:

- Return a table that contains the number of bikeshare rides that begin at each starting station, organized alphabetically by the starting station.
- Return a table that contains the number of bikeshare rides that begin at each starting station, organized numerically from lowest to highest.
- Return a table that contains the number of bikeshare rides that begin at each starting station, organized numerically from highest to lowest.

Each of the commands below is a separate query. For each command:

1. Clear the query **Editor**.
2. Copy and paste the command into the query **Editor**.
3. Click **Run**. Examine the results.

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Open Google Cloud console

**Username**

student-01-a74bdd6d45a3

**Password**

YS1vmA0FWOE3

**Project ID**

qwiklabs-gcp-00-cd593d3

```
SELECT start_station_name, COUNT(*) AS num FROM `bigquery-public-data.london_bicycles.cycle_hire` GROUP BY start_station_name ORDER BY start_station_name;
```

```
SELECT start_station_name, COUNT(*) AS num FROM `bigquery-public-data.london_bicycles.cycle_hire` GROUP BY start_station_name ORDER BY num;
```

```
SELECT start_station_name, COUNT(*) AS num FROM `bigquery-public-data.london_bicycles.cycle_hire` GROUP BY start_station_name ORDER BY num DESC;
```

The results of the last query lists start locations by the number of starts from that location.

You see that "Hyde Park Corner, Hyde Park" has the highest number of starts. However, as a fraction of the total (671688/83434866), you see that < 1% of rides start from this

BigQuery – qwiklabs-gcp-00-cd5

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cycle_hire | *Untitled...ery

Untitled query | Run | Download | Share | Schedule | Open in | More | Save

```
SELECT start_station_name, COUNT(*) AS num FROM `bigquery-public-data.london_bicycles.cycle_hire` GROUP BY
start_station_name ORDER BY start_station_name;
```

Query completed

Using on-demand processing quota

Query results | Save results | Open in

Job information | Results | Visualization | JSON | Execution details | Execution graph

| Row | start_station_name | num |
|---|---|---|
| 1 | Abbey Orchard Street, Westmin... | 121574 |
| 2 | Abbotsbury Road, Holland Park | 38191 |
| 3 | Aberdeen Place, St. John's Wood | 61413 |
| 4 | Aberfeldy Street, Poplar | 21158 |
| 5 | Abingdon Green, Great College ... | 4669 |
| 6 | Abingdon Green, Westminster | 134157 |
| 7 | Abingdon Villas, Kensington | 64312 |
| 8 | Abyssinia Close, Clapham Junc... | 52771 |
| 9 | Ackroyd Drive, Bow | 61193 |
| 10 | Ada Street, Hackney Central | 106514 |
| 11 | Addison Road, Holland Park | 40848 |

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Air: Poor Tomorrow | Search | ENG IN | 05:13 PM 27-11-2025

Untitled query

```
SELECT start_station_name, COUNT(*) AS num FROM `bigquery-public-data.london_bicycles.cycle_hire` GROUP BY start_station_name ORDER BY num;
```

Query completed

Using on-demand processing quota

## Query results

Job information | Results | Visualization | JSON | Execution details | Execution graph

| Row | start_station_name | num |
|---|---|---|
| 1 | PENTON STREET COMMS TEST... | 1 |
| 2 | York Way, Camden | 1 |
| 3 | LSP1 | 2 |
| 4 | One London | 4 |
| 5 | Blackfriars road, Southwark | 5 |
| 6 | LSP2 | 7 |
| 7 | Monier Road | 11 |
| 8 | Monier Road, Newham | 34 |
| 9 | Allington street, Off Victoria Str... | 55 |
| 10 | Contact Centre, Southbury House | 60 |
| 11 | Pop Up Dock 2 | 70 |

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Untitled query

```sql
SELECT start_station_name, COUNT(*) AS num FROM `bigquery-public-data.london_bicycles.cycle_hire` GROUP BY
start_station_name ORDER BY num DESC;
```

Query completed

Using on-demand processing quota

Query results

Job information | Results | Visualization | JSON | Execution details | Execution graph

| Row | start_station_name | num |
|---|---|---|
| 1 | Hyde Park Corner, Hyde Park | 671688 |
| 2 | Belgrove Street , King's Cross | 593065 |
| 3 | Waterloo Station 3, Waterloo | 527112 |
| 4 | Albert Gate, Hyde Park | 461108 |
| 5 | Black Lion Gate, Kensington Gar... | 459716 |
| 6 | Waterloo Station 1, Waterloo | 419334 |
| 7 | Wellington Arch, Hyde Park | 392889 |
| 8 | Hop Exchange, The Borough | 385926 |
| 9 | Wormwood Street, Liverpool Str... | 354663 |
| 10 | Triangle Car Park, Hyde Park | 353033 |
| 11 | Duke Street Hill, London Bridge | 334976 |

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End Lab 00:53:22

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Open Google Cloud console

Username
student-01-a74bdd6d45a3(

Password
YS1vmA0FWOE3

Project ID
qwiklabs-gcp-00-cd593d3:

# Task 4. Working with Cloud SQL

## Exporting queries as CSV files

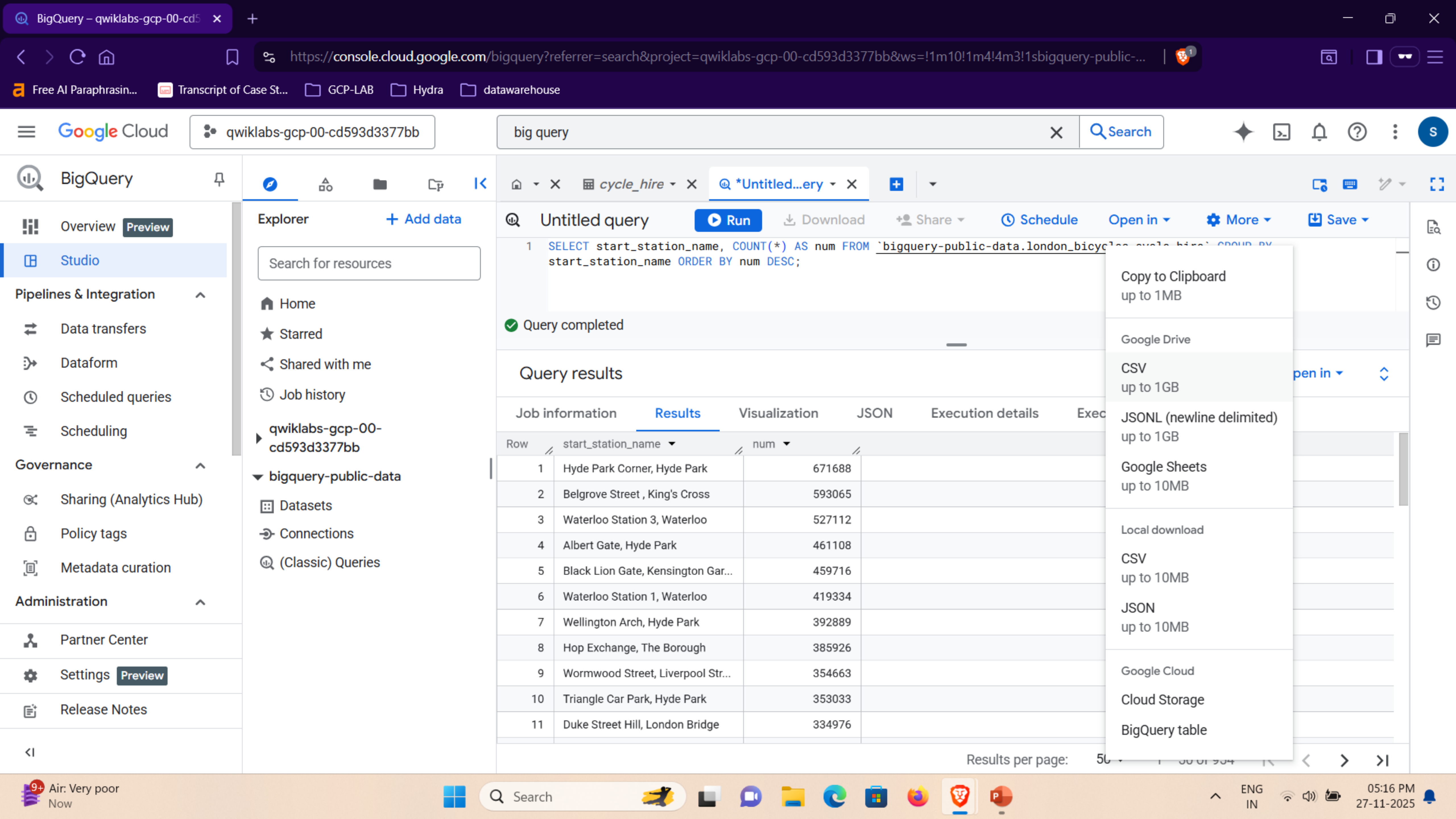
Cloud SQL is a fully-managed database service that makes it easy to set up, maintain, manage, and administer your relational PostgreSQL and MySQL databases in the cloud. There are two formats of data accepted by Cloud SQL: dump files (.sql) or CSV files (.csv). You will learn how to export subsets of the `cycle_hire` table into CSV files and upload them to Cloud Storage as an intermediate location.

Back in the BigQuery Console, this should have been the last command that you ran:

```
SELECT start_station_name, COUNT(*) AS num FROM `bigquery-
public-data.london_bicycles.cycle_hire` GROUP BY
start_station_name ORDER BY num DESC;
```

1. In the Query Results section click **Save results** > **CSV(Local download)**. This initiates a download, which saves this query as a CSV file. Note the location and

0/100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SELECT start_station_name, COUNT(*) AS num FROM `bigquery-public-data.london_bicyc
start_station_name ORDER BY num DESC;
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Query results
Job information
Results
Visualization
JSON
Execution details
Row
start_station_name
num
1 Hyde Park Corner, Hyde Park 671688
2 Belgrove Street , King's Cross 593065
3 Waterloo Station 3, Waterloo 527112
4 Albert Gate, Hyde Park 461108
5 Black Lion Gate, Kensington Gar... 459716
6 Waterloo Station 1, Waterloo 419334
7 Wellington Arch, Hyde Park 392889
8 Hop Exchange, The Borough 385926
9 Wormwood Street, Liverpool Str... 354663
10 Triangle Car Park, Hyde Park 353033
11 Duke Street Hill, London Bridge 334976
Copy to Clipboard
up to 1MB
Google Drive
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up to 1GB
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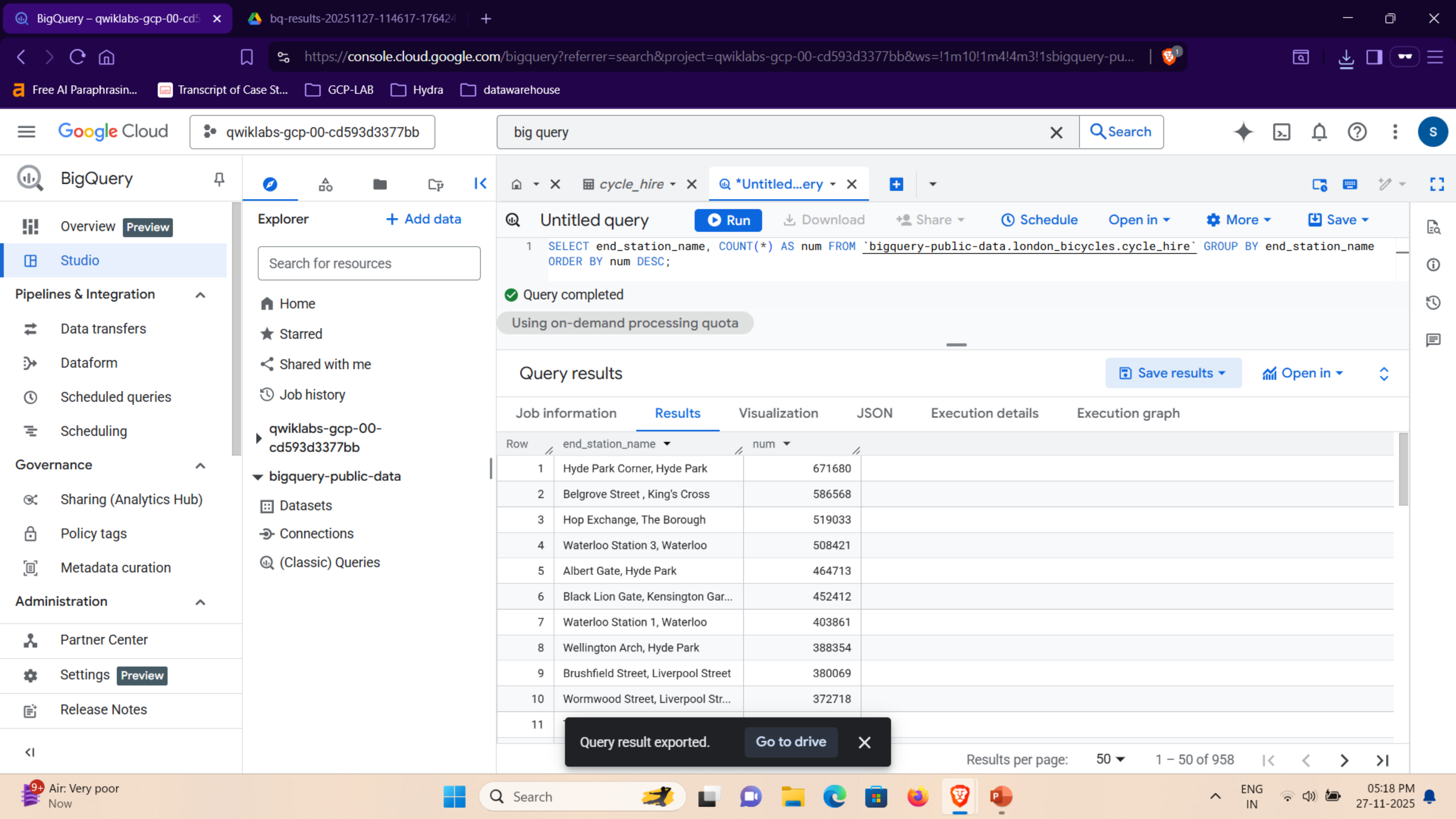

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SELECT end_station_name, COUNT(*) AS num FROM `bigquery-public-data.london_bicycles.cycle_hire` GROUP BY end_station_name
ORDER BY num DESC;
Query completed
Using on-demand processing quota
Query results
Save results
Open in
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Row
end_station_name
num
1 Hyde Park Corner, Hyde Park 671680
2 Belgrove Street , King's Cross 586568
3 Hop Exchange, The Borough 519033
4 Waterloo Station 3, Waterloo 508421
5 Albert Gate, Hyde Park 464713
6 Black Lion Gate, Kensington Gar... 452412
7 Waterloo Station 1, Waterloo 403861
8 Wellington Arch, Hyde Park 388354
9 Brushfield Street, Liverpool Street 380069
10 Wormwood Street, Liverpool Str... 372718
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Open Google Cloud console

Username
student-01-a74bdd6d45a3

Password
YS1vmA0FWOE3

Project ID
qwiklabs-gcp-00-cd593d3

initiates a download, which saves this query as a CSV file. Note the location and the name of this downloaded file—you will need it in the following section.

## Upload CSV files to Cloud Storage

1. Go to the Cloud Console where you'll create a storage bucket where you can upload the files you just created.
2. Select **Navigation menu** > **Cloud Storage** > **Buckets**, and then click **CREATE BUCKET**.

**Note:** If prompted, click **LEAVE** for Unsaved work.

3. Enter a unique name for your bucket, keep all other settings as default, and click **Create**.
4. If prompted, click **Confirm** for `Public access will be prevented` dialog.

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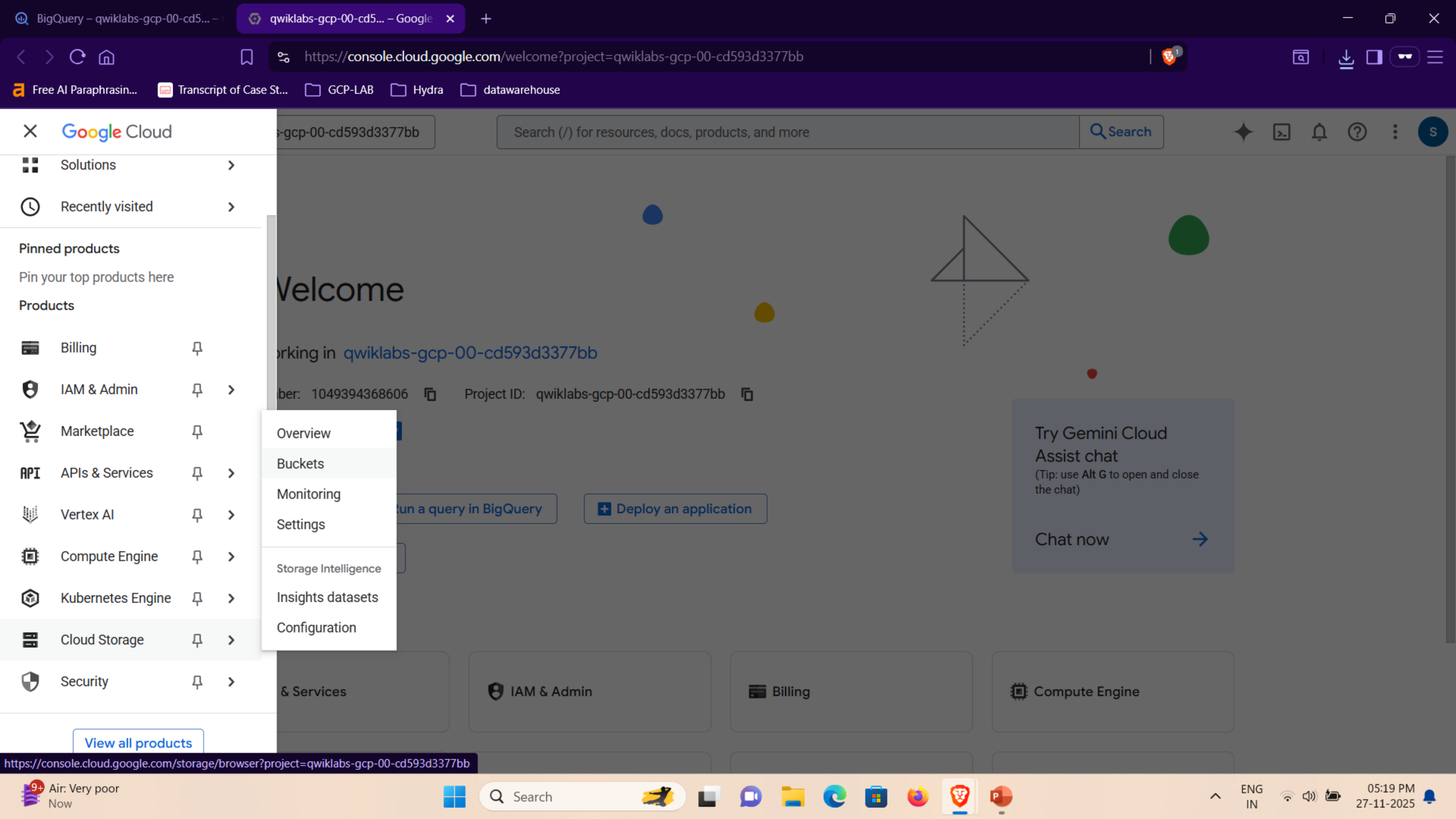
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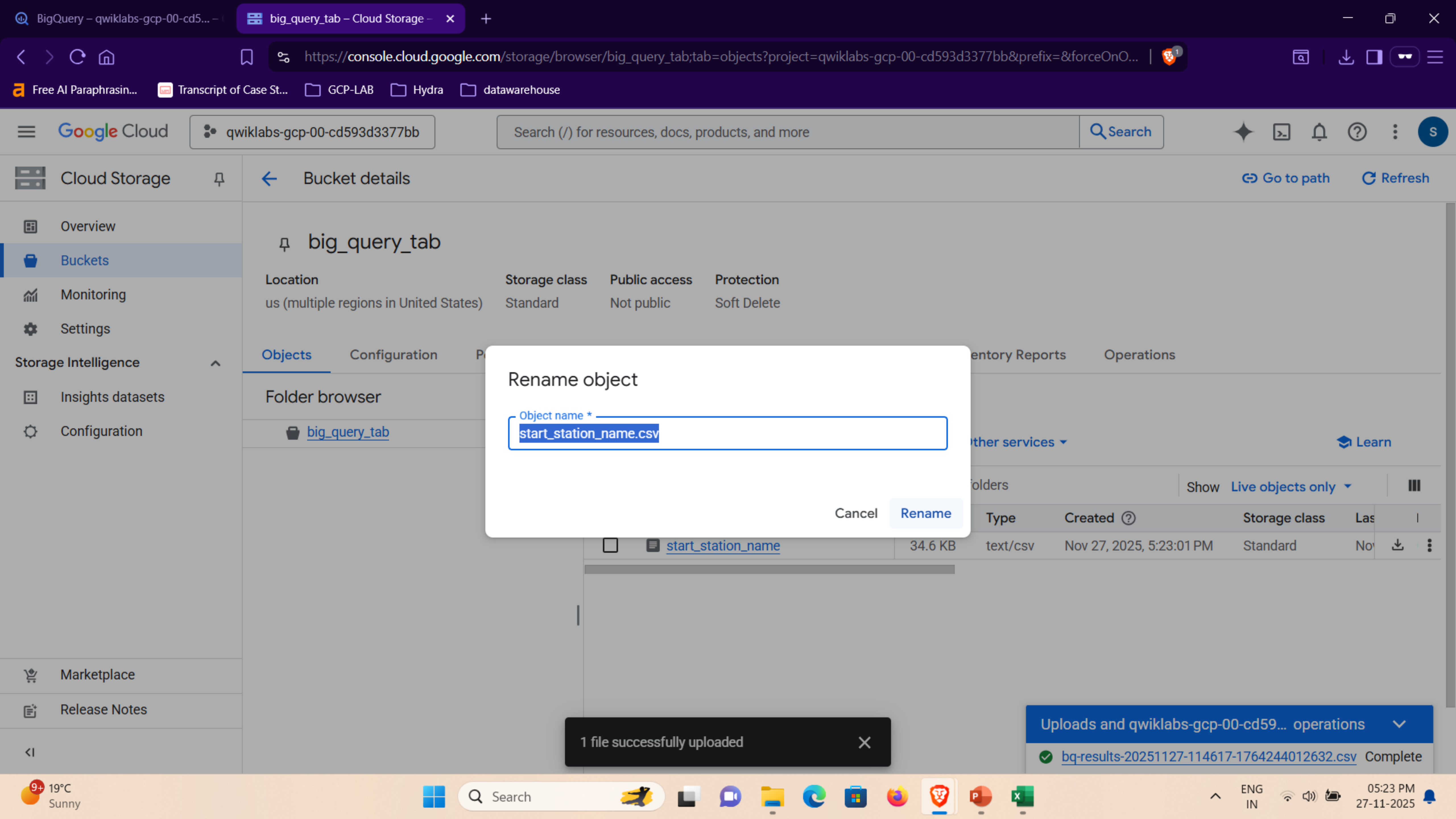
BigQuery – qwiklabs-gcp-00-cd5...
big_query_tab – Cloud Storage –
https://console.cloud.google.com/storage/browser/big_query_tab;tab=objects?project=qwiklabs-gcp-00-cd593d3377bb&prefix=&forceOnO...
Free AI Paraphrasin...
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Hydra
datawarehouse
Google Cloud
qwiklabs-gcp-00-cd593d3377bb
Search (/) for resources, docs, products, and more
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Buckets
Monitoring
Settings
Storage Intelligence
Insights datasets
Configuration
Marketplace
Release Notes
Bucket details
Go to path
Refresh
big_query_tab
Location
us (multiple regions in United States)
Storage class
Standard
Public access
Not public
Protection
Soft Delete
Objects
Configuration
entory Reports
Operations
Folder browser
big_query_tab
Rename object
Object name *
start_station_name.csv
Cancel
Rename
ther services
Learn
olders
Show
Live objects only
Type
Created
Storage class
Las
start_station_name
34.6 KB
text/csv
Nov 27, 2025, 5:23:01 PM
Standard
No
1 file successfully uploaded
Uploads and qwiklabs-gcp-00-cd59... operations
bq-results-20251127-114617-1764244012632.csv
Complete
19°C
Sunny
Search
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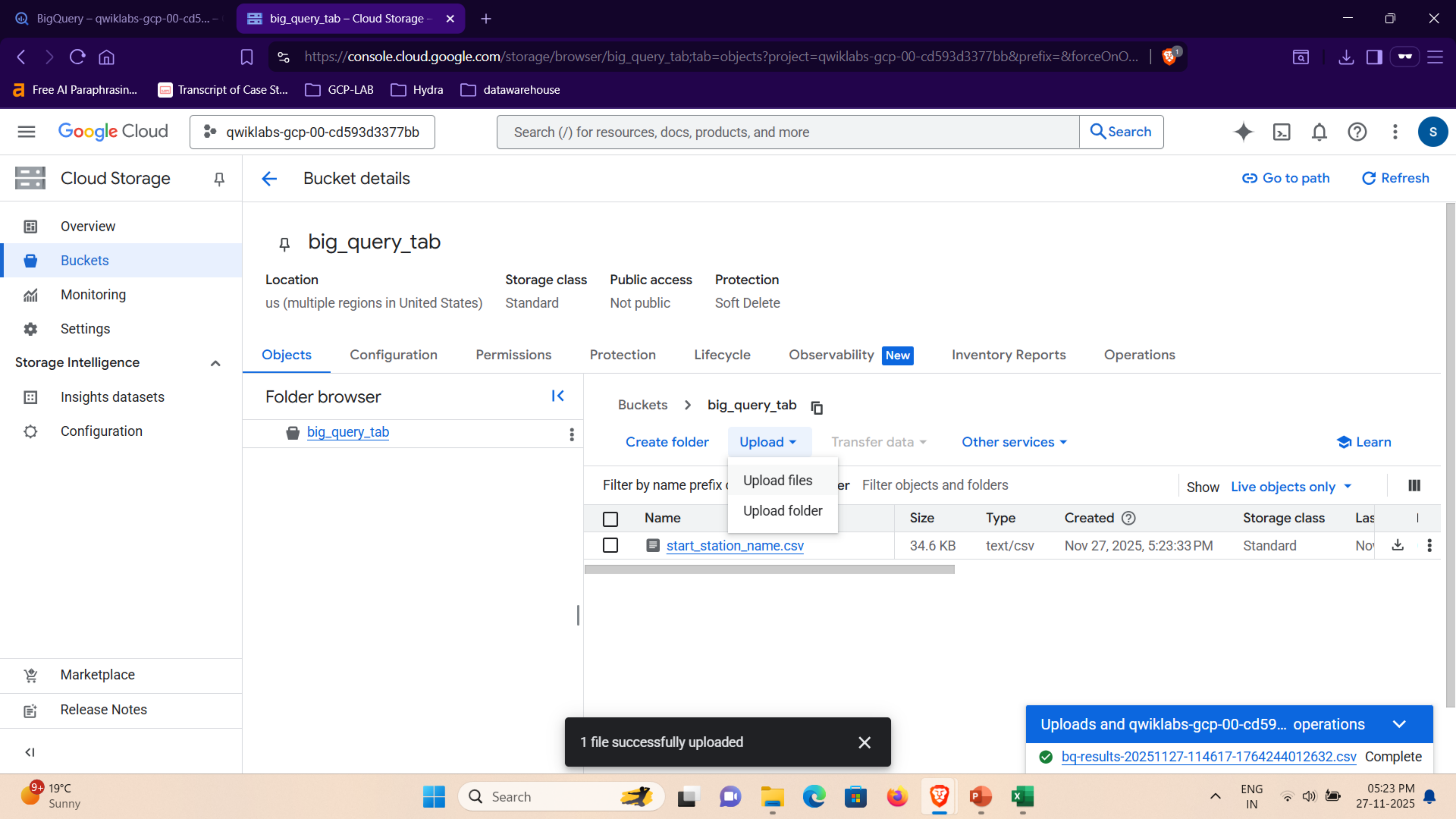
BigQuery – qwiklabs-gcp-00-cd5...
big_query_tab – Cloud Storage –
https://console.cloud.google.com/storage/browser/big_query_tab;tab=objects?project=qwiklabs-gcp-00-cd593d3377bb&prefix=&forceOnO...
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big_query_tab
Location
us (multiple regions in United States)
Storage class
Standard
Public access
Not public
Protection
Soft Delete
Objects
Configuration
Permissions
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Lifecycle
Observability
New
Inventory Reports
Operations
Folder browser
big_query_tab
Buckets
big_query_tab
Create folder
Upload
Transfer data
Other services
Learn
Upload files
Upload folder
Filter by name prefix
Filter objects and folders
Show
Live objects only
Name
Size
Type
Created
Storage class
start_station_name.csv
34.6 KB
text/csv
Nov 27, 2025, 5:23:33 PM
Standard
1 file successfully uploaded
Uploads and qwiklabs-gcp-00-cd59... operations
bq-results-20251127-114617-1764244012632.csv
Complete
19°C
Sunny
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27-11-2025

Lab setup instructions and requirements

End Lab 00:44:25

**Caution:** When you are in the console, do not deviate from the lab instructions. Doing so may cause your account to be blocked. Learn more.

Open Google Cloud console

**Username**

student-01-a74bdd6d45a3(

**Password**

YS1vmA0FWOE3

**Project ID**

qwiklabs-gcp-00-cd593d3:

*Assessment Completed! Buckets: ["big_query_tab"]*

## Task 5. Create a Cloud SQL instance

In the console, select **Navigation menu** > **Cloud SQL**.

1. Click **CREATE INSTANCE > Choose MySQL** .
2. For **Choose a Cloud SQL edition**, select **Enterprise**.
3. For **Instance ID**, type **my-demo**.
4. For **Password**, type *[a secure password]* (and remember it!).
5. For **Database version**, select **MySQL 8**.
6. For **Edition preset**, select **Development** (4 vCPU, 16 GB RAM, 100 GB Storage, Single zone).

25/100

Contents
Introduction to SQL for BigQuery and Cloud SQL

Lab setup instructions and requirements

End Lab 00:40:55

**Caution:** When you are in the console, do not deviate from the lab instructions. Doing so may cause your account to be blocked. Learn more.

Open Google Cloud console

**Username**
student-01-a74bdd6d45a3(

**Password**
YS1vmA0FWOE3

**Project ID**
qwiklabs-gcp-00-cd593d3:

4. For **Password**, type *[a secure password]* (and remember it!).
5. For **Database version**, select **MySQL 8**.
6. For **Edition preset**, select **Development** (4 vCPU, 16 GB RAM, 100 GB Storage, Single zone).

**Warning:** if you choose a preset larger than Development, your project will be flagged and your lab will be terminated.

7. Set the **Region** field as `us-central1`.
8. Set the **Multi zones (Highly available)** > **Primary Zone** field as `us-central1-c`.
9. Click **CREATE INSTANCE**.

**Note:** It might take a few minutes for the instance to be created. Once it is, you will see a green checkmark next to the instance name on the SQL instances page.

50/100

BigQuery – qwiklabs-gcp-00-cd5... big_query_tab – Cloud Storage – qwi qwiklabs-gcp-00-cd5... – Google

https://console.cloud.google.com/welcome?project=qwiklabs-gcp-00-cd593d3377bb

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Google Cloud

-gcp-00-cd593d3377bb

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- APIs & Services
- Vertex AI
- Compute Engine
- Kubernetes Engine
- Cloud Storage
- Security
- BigQuery
- Monitoring
- Cloud Run
- VPC Network
- Cloud SQL

View all products

Get Started

Instances

Backups

Velcome

rking in qwiklabs-gcp-00-cd593d3377bb

ber: 1049394368606 Project ID: qwiklabs-gcp-00-cd593d3377bb

Cloud Hub New

e a VM Run a query in BigQuery Deploy an application

e a storage bucket

Try Gemini Cloud Assist chat

(Tip: use Alt G to open and close the chat)

Chat now

IAM & Admin Billing Compute Engine

https://console.cloud.google.com/sql/get-started?project=qwiklabs-gcp-00-cd593d3377bb

19°C Sunny Search ENG IN 05:25 PM 27-11-2025

Cloud SQL

- Get Started
- Instances
- Backups

## Features and benefits

### Highly available, protected data

Recover data from any point in time. Safeguard your data with automated backups. Utilize greater than 99.99% availability. Replicate across regions.

Learn more about MySQL, PostgreSQL, and SQL Server

### Pricing and scalability for any workload

Size your instance to fit your needs, with per-second billing and databases that are easily stopped and restarted. Add up to 128 vCPUs and up to 864 GB of RAM and 64 TB of storage. Create read replicas to handle increasing read traffic.

Learn more about Cloud SQL pricing

### Minimize downtime and stay up-to-date

Get notified of upcoming maintenance, and reschedule or deny it. Stay current with automatic minor version upgrades, and in-place major database version upgrades.

Learn more about maintenance controls for MySQL, PostgreSQL, and SQL Server

### Integrate with the best of cloud

Connect from App Engine, Compute Engine, Kubernetes, and more. Analyze data using federated queries from BigQuery.

Learn more about federated queries from Big Query, Compute Engine, and Kubernetes

### Simple data migration

Launch a guided experience using Database Migration to move workloads from on-premises or other cloud providers. Continuous replication minimizes downtime.

Learn more about Database Migration

### Insights and recommendations

Understand and resolve performance issues with pre-built dashboards, visual query plans, and robust database metrics. Receive tips to reduce cost and improve performance and security.

Learn more about recommendations and insights for MySQL and PostgreSQL

## Get started

Create instance

View documentation

Release Notes

BigQuery – qwiklabs-gcp-00-cd5... | big_query_tab – Cloud Storage – qwi | Create an instance – Cloud SQL

https://console.cloud.google.com/sql/choose-instance-engine?project=qwiklabs-gcp-00-cd593d3377bb

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Google Cloud | qwiklabs-gcp-00-cd593d3377bb | Search (/) for resources, docs, products, and more | Search

Cloud SQL

- Get Started
- Instances
- Backups

Cloud SQL ← Create an instance

## Choose your database engine

### MySQL

Versions: 8.4, 8.0, 5.7, 5.6

Choose MySQL

### PostgreSQL

Versions: 18, 17, 16, 15, 14, 13, 12, 11, 10, 9.6

Choose PostgreSQL

### SQL Server

Versions: 2022, 2019, 2017

Choose SQL Server

Want more context on the Cloud SQL database engines? Learn more

### Introducing AlloyDB for PostgreSQL

PostgreSQL plus the best of the cloud: scale-out storage and compute, intelligent caching, and AI/ML-powered management. Choose AlloyDB for 4x faster transactional workloads, more efficient analytical and vector queries, and an industry-leading 99.99% SLA, inclusive of maintenance.

Free option available

Choose AlloyDB | Learn more

Release Notes

https://console.cloud.google.com/sql/instances/create;engine=MySQL?project=qwiklabs-gc...

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Cloud SQL – qwiklabs-gcp-00-0b

https://console.cloud.google.com/sql/instances/create;engine=MySQL?project=qwiklabs-gcp-00-0b0924b5d37c

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Google Cloud qwiklabs-gcp-00-0b0924b5d37c cloud sql Search

Cloud SQL

- Get Started
- Instances
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Create a MySQL instance **Note:** Choose Enterprise only, if other big edition is selected and used the lab will end automatically

## Choose a Cloud SQL edition

A Cloud SQL edition determines foundational characteristics of your instance. Choose the best option for your price and performance needs. Learn more

**Enterprise Plus**

- 99.99% availability SLA
- Sub-second planned maintenance downtime
- Near-zero downtime instance scale-up
- Performance optimized machines
- Up to 35 days point-in-time recovery window
- Up to 3x higher read throughput with data cache
- Advanced disaster recovery with easy switchback
- Support for managed connection pool

**Enterprise** (selected)

- 99.95% availability SLA
- Less than 60 seconds planned maintenance downtime
- General purpose machines
- Up to 7 days point-in-time recovery window

Choose a preset for this edition. Presets can be customized later as needed.

## Summary

| | |
|---|---|
| Cloud SQL Edition | Enterprise |
| Region | us-central1 (Iowa) |
| DB Version | MySQL 8.0 |
| Machine type | db-custom-8-32768 |
| vCPUs | 8 vCPU |
| RAM | 32 GB |
| Data Cache | Disabled |
| Storage | 250 GB SSD |
| Connections | Public IP |
| Backup | Automated |
| Availability | Multiple zones (Highly available) |
| Point-in-time recovery | Enabled |
| Network throughput (MB/s) | 2,000 of 2,000 |
| IOPS | Read: 13,500 of 15,000<br>Write: 13,500 of 15,000 |
| Disk throughput (MB/s) | Read: 120.0 of 800.0<br>Write: 120.0 of 800.0 |

## Pricing estimate (without discounts)

These items represent Cloud

Release Notes

19°C Sunny Search ENG IN 05:50 PM 27-11-2025

BigQuery – qwiklabs-gcp-00-cd5... | big_query_tab – Cloud Storage – qwi | Cloud SQL – qwiklabs-gcp-00-cd

https://console.cloud.google.com/sql/instances/create;engine=MySQL?project=qwiklabs-gcp-00-cd593d3377bb

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Google Cloud | qwiklabs-gcp-00-cd593d3377bb | Search (/) for resources, docs, products, and more | Search

Cloud SQL

- Get Started
- Instances
- Backups

Release Notes

# Create a MySQL instance

Edition preset
Production

Compare edition presets

## Instance info

Database version *
MySQL 8.0

Show minor versions

Instance ID *
my-demo

Use lowercase letters, numbers, and hyphens. Start with a letter.

Password *
2fA*mLdNXIP`nl[B
Generate

Set a password for the root user. Learn more

No password

- Must contain an uppercase letter, lowercase letter, number and non-alphanumeric character
- Must not contain the username
- Must be at least 8 characters long

Password policy

## Summary

| | |
|---|---|
| Cloud SQL Edition | Enterprise Plus |
| Region | us-central1 (Iowa) |
| DB Version | MySQL 8.0 |
| Machine type | db-perf-optimized-N-8 |
| vCPUs | 8 vCPU |
| RAM | 64 GB |
| Data Cache | Enabled (375 GB) |
| Storage | 250 GB SSD |
| Connections | Public IP |
| Backup | Automated |
| Availability | Multiple zones (Highly available) |
| Point-in-time recovery | Enabled |
| Network throughput (MB/s) | 2,000 of 2,000 |
| IOPS | Read: 13,500 of 15,000<br>Write: 13,500 of 15,000 |
| Disk throughput (MB/s) | Read: 120.0 of 800.0<br>Write: 120.0 of 800.0 |

## Pricing estimate (without discounts)

These items represent Cloud SQL compute, memory and

19°C Sunny | Search | ENG IN | 05:27 PM 27-11-2025

# Create a MySQL instance

connection pool

Choose a preset for this edition. Presets can be customized later as needed.

Edition preset
Development

Compare edition presets

## Instance info

Database version *
MySQL 8.0

Show minor versions

Instance ID *
my-demo

Use lowercase letters, numbers, and hyphens. Start with a letter.

Password *
2fA*mLdNXIP`nl[B
Generate

Set a password for the root user. Learn more

No password

Must contain an uppercase letter, lowercase let
alphanumeric character

"Development" configuration applied

## Summary

| | |
|---|---|
| Cloud SQL Edition | Enterprise Plus |
| Region | us-central1 (Iowa) |
| DB Version | MySQL 8.0 |
| Machine type | db-perf-optimized-N-4 |
| vCPUs | 4 vCPU |
| RAM | 32 GB |
| Data Cache | Enabled (375 GB) |
| Storage | 250 GB SSD |
| Connections | Public IP |
| Backup | Automated |
| Availability | Single zone |
| Point-in-time recovery | Enabled |
| Network throughput (MB/s) | 1,000 of 1,000 |
| IOPS | Read: 13,500 of 15,000<br>Write: 13,500 of 15,000 |
| Disk throughput (MB/s) | Read: 120.0 of 240.0<br>Write: 120.0 of 240.0 |

## ing estimate (without discounts)

These items represent Cloud SQL compute, memory and

BigQuery – qwiklabs-gcp-00-cd5... | big_query_tab – Cloud Storage – qwi | Cloud SQL – qwiklabs-gcp-00-cd

https://console.cloud.google.com/sql/instances/create;engine=MySQL?project=qwiklabs-gcp-00-cd593d3377bb

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Cloud SQL

- Get Started
- Instances
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Release Notes

Create a MySQL instance

## Choose region and zonal availability

For better performance, keep your data close to the services that need it. Region is permanent, while zone can be changed any time.

You have the resource location org policy in effect that is restricting your regions.

Region

us-central1 (Iowa)

Zonal availability

- Single zone
  In case of outage, no failover. Not recommended for production.
- Multiple zones (Highly available)
  Automatic failover to another zone within your selected region. Recommended for production instances. Increases cost.

Primary zone: us-central1-c

Secondary zone: Any (different from pri...

Hide zones

## Customize your instance

You can also customize instance configurations later

| | |
|---|---|
| Disk throughput (MB/s) | Read: 120.0 of 240.0<br>Write: 120.0 of 240.0 |

## Pricing estimate (without discounts)

These items represent Cloud SQL compute, memory and storage resources only, and reflect how you configured your instance so far. Discounts not included in estimate. Learn more

| Item | Hourly cost (estimate) |
|---|---|
| 4 vCPU ($0.054 per vCPU/hour) | $0.21 |
| 32 GiB RAM ($0.009 per GiB/hour) | $0.29 |
| 250 GiB SSD ($0.17 per GiB/month) | $0.06 |
| 375 GiB data cache ($0.16 per GiB/month) | $0.08 |
| Standby VM (high availability) | $0.65 |
| **Total without usage discounts** | **$1.29** |

### Usage and traffic costs

These costs vary based on your feature usage, traffic, or data location. Learn more

Standard backups ($0.08 per GiB)

Enhanced Backups ($0.10 per GiB)
Enhanced backups, managed by the Backup and DR Service,

19°C Sunny | Search | ENG IN | 05:30 PM 27-11-2025

my-demo – Overview – Cloud SQL

https://console.cloud.google.com/sql/instances/my-demo/overview?project=qwiklabs-gcp-00-0b0924b5d37c

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Google Cloud | qwiklabs-gcp-00-0b0924b5d37c | cloud sql | Search

Cloud SQL

Overview | Edit | Import | Export | Restart | Stop | Delete | Clone | Failover

Primary instance

- Overview
- Cloud SQL Studio
- System insights
- Query insights
- Connections
- Users
- Databases
- Backups
- Replicas
- Operations

Release Notes

All instances > my-demo

# my-demo

MySQL 8.0

## Learn the basics of Cloud SQL

0 of 3 completed

1. Import data
2. Query and explore data
3. Connect source application

With your Cloud SQL instance now running, the next step is to import data. You can import CSV and SQL files from your local file system or from a GCS bucket.

If you don't have existing data, you can create a database, then create a table in Cloud SQL Studio in the next step.

Import data | Mark as complete

1 hour | 6 hours | 1 day | 7 days | 30 days | Custom

Chart

CPU utilization

No data is available for the selected time frame.

UTC+5:30 | 8:00 PM | 10:00 PM | Nov 27 | 2:00 AM | 4:00 AM | 6:00 AM | 8:00 AM | 10:00 AM

**Uploads and qwiklabs-gcp-00-0b09... operations**

Created my-demo. 5:58:59 PM GMT+5

Uploading 2 files

| | |
|---|---|
| end_station_data.csv | Complete |
| start_station_name.csv | Complete |

Trending videos
Stranger Things...

Search

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Instances – Cloud SQL – qwiklabs

https://console.cloud.google.com/sql/instances?project=qwiklabs-gcp-00-0b0924b5d37c

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Google Cloud | qwiklabs-gcp-00-0b0924b5d37c | cloud sql | Search

Cloud SQL

- Get Started
- Instances
- Backups

Instances | Create Instance | Migrate Database | Hide info panel | Learn

Starting Feb 1, 2025, all instances running community end-of-life versions of PostgreSQL and MySQL are under extended support. These instances will be charged for extended support from May 1, 2025. Upgrade your instances running end-of-life versions before May 1, 2025 to prevent additional charges. Learn more

View affected instances | Dismiss

Filter Enter property name or value

| Status | Instance ID | Issues | Cloud SQL edition | Type | Public IP address | Actions |
|---|---|---|---|---|---|---|
| ✓ | my-demo | | Enterprise | MySQL 8.0 | 136.119.82.6 | ⋮ |

No instances selected

Monitoring | Labels

No instances selected.

Release Notes

18°C Clear | Search | ENG IN | 06:00 PM 27-11-2025

Lab setup instructions and requirements

End Lab 00:52:23

**Caution:** When you are in the console, do not deviate from the lab instructions. Doing so may cause your account to be blocked. Learn more.

Open Google Cloud console

**Username**

student-00-d949fb26ec6c(

**Password**

8pbrEyfTWRka

**Project ID**

qwiklabs-gcp-00-0b0924b!

# Task 6. New queries in Cloud SQL

## CREATE keyword (databases and tables)

Now that you have a Cloud SQL instance up and running, create a database inside of it using the Cloud Shell Command Line.

1. Open Cloud Shell by clicking on the icon in the top right corner of the console.
2. Run the following command to set your project ID as an environment variable:

```
export PROJECT_ID=$(gcloud config get-value project)
gcloud config set project $PROJECT_ID
```

## Create a database in Cloud Shell

1. Run the following command in Cloud Shell to setup auth without opening up a

Instances – Cloud SQL – qwiklabs

https://console.cloud.google.com/sql/instances?project=qwiklabs-gcp-00-0b0924b5d37c&cloudshell=true

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Google Cloud qwiklabs-gcp-00-0b0924b5d37c cloud sql Search

Cloud SQL Instances Create Instance Migrate Database Hide info panel Learn

CLOUD SHELL
Terminal (qwiklabs-gcp-00-0b0924b5d37c) Open Editor

Gemini CLI is available in Cloud Shell terminal! Type gemini to try it. Learn more Don't show again Dismiss

```
Welcome to Cloud Shell! Type "help" to get started, or type "gemini" to try prompting with Gemini CLI.
Your Cloud Platform project in this session is set to qwiklabs-gcp-00-0b0924b5d37c.
Use `gcloud config set project [PROJECT_ID]` to change to a different project.
student_00_d949fb26ec6c@cloudshell:~ (qwiklabs-gcp-00-0b0924b5d37c)$ export PROJECT_ID=$(gcloud config get-value project)
gcloud config set project $PROJECT_ID
Your active configuration is: [cloudshell-2292]
INFORMATION: Project 'qwiklabs-gcp-00-0b0924b5d37c' has no 'environment' tag set. Use either 'Production', 'Development', 'Test', or 'Staging'. Add an 'environment' tag using `gcloud resource
-manager tags bindings create`.
Updated property [core/project].
student_00_d949fb26ec6c@cloudshell:~ (qwiklabs-gcp-00-0b0924b5d37c)$
```

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Introduction to SQL for BigQuery and Cloud SQL

Lab setup instructions and requirements

End Lab 00:51:03

**Caution:** When you are in the console, do not deviate from the lab instructions. Doing so may cause your account to be blocked. Learn more.

Open Google Cloud console

Username
student-00-d949fb26ec6c(

Password
8pbrEyfTWRka

Project ID
qwiklabs-gcp-00-0b0924b!

```
gcloud config set project $PROJECT_ID
```

## Create a database in Cloud Shell

1. Run the following command in Cloud Shell to setup auth without opening up a browser.

```
gcloud auth login --no-launch-browser
```

If prompted [Y/n], press **Y** and then **ENTER**.

This will give you a link to open in your browser. Open the link in the same browser where you are logged in to the qwiklabs account. Once you login you will get a verification code to copy. Paste that code in the cloud shell.

2. Run the following command to connect to your SQL instance, replacing `my-demo` if you used a different name for your instance:

75/100

Instances – Cloud SQL – qwiklabs-gcp | gcloud CLI Remote Login

https://sdk.cloud.google.com/authcode.html?state=197FHQcEg2L5CWVJhxDRlTLxz4us5C&code=4/0Ab32j90kGtw-zFEkcxfnpFnDDco-T5TMeLGbK...

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# Sign in to the gcloud CLI

You are seeing this page because you ran the following command in the gcloud CLI from this or another machine. If this is not the case, close this tab.

```
gcloud auth login --no-launch-browser
```

Enter the following verification code in gcloud CLI on the machine you want to log into. This is a credential **similar to your password** and should not be shared with others.

```
4/0Ab32j90kGtw-zFEkcxfnpFnDDco-T5TMeLG
bKq1V4-GU17bWHbY7oVRTd4uiwnWJXcl49g
```

Copy

You can close this tab when you're done.

18°C
Clear

Search

ENG
IN

06:03 PM
27-11-2025

Instances – Cloud SQL – qwiklab | gcloud CLI Remote Login

https://console.cloud.google.com/sql/instances?project=qwiklabs-gcp-00-0b0924b5d37c&cloudshell=true

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Google Cloud | qwiklabs-gcp-00-0b0924b5d37c | cloud sql | Search

Cloud SQL | Instances | Create Instance | Migrate Database | Hide info panel | Learn

CLOUD SHELL
Terminal (qwiklabs-gcp-00-0b0924b5d37c) | Open Editor

Gemini CLI is available in Cloud Shell terminal! Type gemini to try it. Learn more | Don't show again | Dismiss

```
Welcome to Cloud Shell! Type "help" to get started, or type "gemini" to try prompting with Gemini CLI.
Your Cloud Platform project in this session is set to qwiklabs-gcp-00-0b0924b5d37c.
Use `gcloud config set project [PROJECT_ID]` to change to a different project.
student_00_d949fb26ec6c@cloudshell:~ (qwiklabs-gcp-00-0b0924b5d37c)$ export PROJECT_ID=$(gcloud config get-value project)
gcloud config set project $PROJECT_ID
Your active configuration is: [cloudshell-2292]
INFORMATION: Project 'qwiklabs-gcp-00-0b0924b5d37c' has no 'environment' tag set. Use either 'Production', 'Development', 'Test', or 'Staging'. Add an 'environment' tag using `gcloud resource
-manager tags bindings create`.
Updated property [core/project].
student_00_d949fb26ec6c@cloudshell:~ (qwiklabs-gcp-00-0b0924b5d37c)$ gcloud auth login --no-launch-browser

You are already authenticated with gcloud when running
inside the Cloud Shell and so do not need to run this
command. Do you wish to proceed anyway?

Do you want to continue (Y/n)?  Y

Go to the following link in your browser, and complete the sign-in prompts:

    https://accounts.google.com/o/oauth2/auth?response_type=code&client_id=32555940559.apps.googleusercontent.com&redirect_uri=https%3A%2F%2Fsdk.cloud.google.com%2Fauthcode.html&scope=openid+
https%3A%2F%2Fwww.googleapis.com%2Fauth%2Fuserinfo.email+https%3A%2F%2Fwww.googleapis.com%2Fauth%2Fcloud-platform+https%3A%2F%2Fwww.googleapis.com%2Fauth%2Fappengine.admin+https%3A%2F%2Fwww.g
oogleapis.com%2Fauth%2Fsqlservice.login+https%3A%2F%2Fwww.googleapis.com%2Fauth%2Fcompute+https%3A%2F%2Fwww.googleapis.com%2Fauth%2Faccounts.reauth&state=197FHQcEg2L5CWVJhxDRlTLxz4us5C&prompt
=consent&token_usage=remote&access_type=offline&code_challenge=zg60Q2LxRPJNXb5GOYhkRfg9-ml_zGnPASyXYCnFwgM&code_challenge_method=S256

Once finished, enter the verification code provided in your browser: 4/0Ab32j90kGtw-zFEkcxfnpFnDDco-T5TMeLGbKq1V4-GU17bWHbY7oVRTd4uiwnWJXcl49g

You are now logged in as [student-00-d949fb26ec6c@qwiklabs.net].
Your current project is [qwiklabs-gcp-00-0b0924b5d37c].  You can change this setting by running:
  $ gcloud config set project PROJECT_ID
student_00_d949fb26ec6c@cloudshell:~ (qwiklabs-gcp-00-0b0924b5d37c)$
```

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Lab setup instructions and requirements

End Lab 00:49:42

**Caution:** When you are in the console, do not deviate from the lab instructions. Doing so may cause your account to be blocked. Learn more.

Open Google Cloud console

Username
student-00-d949fb26ec6c(

Password
8pbrEyfTWRka

Project ID
qwiklabs-gcp-00-0b0924b!

This will give you a link to open in your browser. Open the link in the same browser where you are logged in to the qwiklabs account. Once you login you will get a verification code to copy. Paste that code in the cloud shell.

2. Run the following command to connect to your SQL instance, replacing `my-demo` if you used a different name for your instance:

```
gcloud sql connect my-demo --user=root --quiet
```

**Note:** It may take a minute to connect to your instance. If you get an message that "Operation failed because another operation was already in progress", you need to wait for the SQL instance to finish being created, and then try to connect again.

3. When prompted, enter the root password you set for the instance. **Note:** The cursor will not move.

You should see similar output:

```
Welcome to the MySQL monitor.  Commands end with ; or \g.
```

75/100

Instances – Cloud SQL – qwiklabs × gcloud CLI Remote Login +

https://console.cloud.google.com/sql/instances?project=qwiklabs-gcp-00-0b0924b5d37c&cloudshell=true

Free AI Paraphrasin... Transcript of Case St... GCP-LAB Hydra datawarehouse

Google Cloud qwiklabs-gcp-00-0b0924b5d37c cloud sql Search

Cloud SQL Instances Create Instance Migrate Database Hide info panel Learn

CLOUD SHELL
Terminal (qwiklabs-gcp-00-0b0924b5d37c) × + Open Editor

Gemini CLI is available in Cloud Shell terminal! Type gemini to try it. Learn more Don't show again Dismiss

```
command. Do you wish to proceed anyway?

Do you want to continue (Y/n)?  Y

Go to the following link in your browser, and complete the sign-in prompts:

    https://accounts.google.com/o/oauth2/auth?response_type=code&client_id=32555940559.apps.googleusercontent.com&redirect_uri=https%3A%2F%2Fsdk.cloud.google.com%2Fauthcode.html&scope=openid+
https%3A%2F%2Fwww.googleapis.com%2Fauth%2Fuserinfo.email+https%3A%2F%2Fwww.googleapis.com%2Fauth%2Fcloud-platform+https%3A%2F%2Fwww.googleapis.com%2Fauth%2Fappengine.admin+https%3A%2F%2Fwww.g
oogleapis.com%2Fauth%2Fsqlservice.login+https%3A%2F%2Fwww.googleapis.com%2Fauth%2Fcompute+https%3A%2F%2Fwww.googleapis.com%2Fauth%2Faccounts.reauth&state=197FHQcEg2L5CWVJhxDRlTLxz4us5C&prompt
=consent&token_usage=remote&access_type=offline&code_challenge=zg60Q2LxRPJNXb5GOYhkRfg9-ml_zGnPASyXYCnFwgM&code_challenge_method=S256

Once finished, enter the verification code provided in your browser: 4/0Ab32j90kGtw-zFEkcxfnpFnDDco-T5TMeLGbKq1V4-GU17bWHbY7oVRTd4uiwnWJXcl49g

You are now logged in as [student-00-d949fb26ec6c@qwiklabs.net].
Your current project is [qwiklabs-gcp-00-0b0924b5d37c].  You can change this setting by running:
  $ gcloud config set project PROJECT_ID
student_00_d949fb26ec6c@cloudshell:~ (qwiklabs-gcp-00-0b0924b5d37c)$ gcloud sql connect my-demo --user=root --quiet
Allowlisting your IP for incoming connection for 5 minutes...done.
Connecting to database with SQL user [root].Enter password:
Welcome to the MySQL monitor.  Commands end with ; or \g.
Your MySQL connection id is 75
Server version: 8.0.41-google (Google)

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owners.

Type 'help;' or '\h' for help. Type '\c' to clear the current input statement.

mysql>
```

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Contents

Introduction to SQL for BigQuery and Cloud SQL

Lab setup instructions and requirements

End Lab 00:47:19

**Caution:** When you are in the console, do not deviate from the lab instructions. Doing so may cause your account to be blocked. Learn more.

Open Google Cloud console

Username

student-00-d949fb26ec6c

Password

8pbrEyfTWRka

Project ID

qwiklabs-gcp-00-0b0924b

```
Type 'help;' or '\h' for help. Type '\c' to clear the current input
statement.

mysql>
```

A Cloud SQL instance comes with pre-configured databases, but you will create your own to store the London bikeshare data.

4. Run the following command at the MySQL server prompt to create a database called `bike`:

```
CREATE DATABASE bike;
```

You should receive the following output:

```
Query OK, 1 row affected (0.05 sec)

mysql>
```

75/100

Instances – Cloud SQL – qwiklabs | gcloud CLI Remote Login

https://console.cloud.google.com/sql/instances?project=qwiklabs-gcp-00-0b0924b5d37c&cloudshell=true

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Google Cloud | qwiklabs-gcp-00-0b0924b5d37c | cloud sql | Search

Cloud SQL | Instances | Create Instance | Migrate Database | Hide info panel | Learn

CLOUD SHELL
Terminal (qwiklabs-gcp-00-0b0924b5d37c) | Open Editor

Gemini CLI is available in Cloud Shell terminal! Type `gemini` to try it. Learn more | Don't show again | Dismiss

```
Go to the following link in your browser, and complete the sign-in prompts:

    https://accounts.google.com/o/oauth2/auth?response_type=code&client_id=32555940559.apps.googleusercontent.com&redirect_uri=https%3A%2F%2Fsdk.cloud.google.com%2Fauthcode.html&scope=openid+
https%3A%2F%2Fwww.googleapis.com%2Fauth%2Fuserinfo.email+https%3A%2F%2Fwww.googleapis.com%2Fauth%2Fcloud-platform+https%3A%2F%2Fwww.googleapis.com%2Fauth%2Fappengine.admin+https%3A%2F%2Fwww.g
oogleapis.com%2Fauth%2Fsqlservice.login+https%3A%2F%2Fwww.googleapis.com%2Fauth%2Fcompute+https%3A%2F%2Fwww.googleapis.com%2Fauth%2Faccounts.reauth&state=197FHQcEg2L5CWVJhxDRlTLxz4us5C&prompt
=consent&token_usage=remote&access_type=offline&code_challenge=zg60Q2LxRPJNXb5GOYhkRfg9-ml_zGnPASyXYCnFwgM&code_challenge_method=S256

Once finished, enter the verification code provided in your browser: 4/0Ab32j90kGtw-zFEkcxfnpFnDDco-T5TMeLGbKq1V4-GU17bWHbY7oVRTd4uiwnWJXcl49g

You are now logged in as [student-00-d949fb26ec6c@qwiklabs.net].
Your current project is [qwiklabs-gcp-00-0b0924b5d37c].  You can change this setting by running:
  $ gcloud config set project PROJECT_ID
student_00_d949fb26ec6c@cloudshell:~ (qwiklabs-gcp-00-0b0924b5d37c)$ gcloud sql connect my-demo --user=root --quiet
Allowlisting your IP for incoming connection for 5 minutes...done.
Connecting to database with SQL user [root].Enter password:
Welcome to the MySQL monitor.  Commands end with ; or \g.
Your MySQL connection id is 75
Server version: 8.0.41-google (Google)

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owners.

Type 'help;' or '\h' for help. Type '\c' to clear the current input statement.

mysql> CREATE DATABASE bike;
Query OK, 1 row affected (0.22 sec)

mysql>
```

18°C Clear | Search | ENG IN | 06:05 PM 27-11-2025

Contents
Introduction to SQL for BigQuery and Cloud SQL

Lab setup instructions and requirements

End Lab 00:46:36

**Caution:** When you are in the console, do not deviate from the lab instructions. Doing so may cause your account to be blocked. Learn more.

Open Google Cloud console

Username
student-00-d949fb26ec6c(

Password
8pbrEyfTWRka

Project ID
qwiklabs-gcp-00-0b0924b!

## Create a table in Cloud Shell

1. Make a table inside of the bike database by running the following command:

```
USE bike;
CREATE TABLE london1 (start_station_name VARCHAR(255), num INT);
```

This statement uses the `CREATE` keyword, but this time it uses the `TABLE` clause to specify that it wants to build a table instead of a database. The `USE` keyword specifies a database that you want to connect to. You now have a table named "london1" that contains two columns, "start_station_name" and "num". `VARCHAR(255)` specifies variable length string column that can hold up to 255 characters and `INT` is a column of type integer.

2. Create another table named "london2" by running the following command:

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Open Google Cloud console

Username

student-00-d949fb26ec6c(

Password

8pbrEyfTWRka

Project ID

qwiklabs-gcp-00-0b0924b!

contains two columns, "start_station_name" and "num". `VARCHAR(255)` specifies variable length string column that can hold up to 255 characters and `INT` is a column of type integer.

2. Create another table named "london2" by running the following command:

Copied!

```
USE bike;
CREATE TABLE london2 (end_station_name VARCHAR(255), num INT);
```

3. Now confirm that your empty tables were created. Run the following commands at the MySQL server prompt:

```
SELECT * FROM london1;
SELECT * FROM london2;
```

You should receive the following output for both commands:

```
Empty set (0.04 sec)
```

Instances – Cloud SQL – qwiklabs | gcloud CLI Remote Login

https://console.cloud.google.com/sql/instances?project=qwiklabs-gcp-00-0b0924b5d37c&cloudshell=true

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CLOUD SHELL
Terminal (qwiklabs-gcp-00-0b0924b5d37c)

Open Editor

Gemini CLI is available in Cloud Shell terminal! Type gemini to try it. Learn more | Don't show again | Dismiss

```
Server version: 8.0.41-google (Google)

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affiliates. Other names may be trademarks of their respective
owners.

Type 'help;' or '\h' for help. Type '\c' to clear the current input statement.

mysql> CREATE DATABASE bike;
Query OK, 1 row affected (0.22 sec)

mysql> USE bike;
Database changed
mysql> CREATE TABLE london1 (start_station_name VARCHAR(255), num INT);
Query OK, 0 rows affected (0.27 sec)

mysql> CREATE TABLE london1 (start_station_name VARCHAR(255), num INT);
ERROR 1050 (42S01): Table 'london1' already exists
mysql> USE bike;
Database changed
mysql> CREATE TABLE london2 (end_station_name VARCHAR(255), num INT);
Query OK, 0 rows affected (0.27 sec)

mysql> SELECT * FROM london1;
Empty set (0.20 sec)

mysql> SELECT * FROM london2;
Empty set (0.21 sec)

mysql>
```

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Contents

Introduction to SQL for BigQuery and Cloud SQL

Lab setup instructions and requirements

End Lab 00:44:35

**Caution:** When you are in the console, do not deviate from the lab instructions. Doing so may cause your account to be blocked. Learn more.

Open Google Cloud console

Username

student-00-d949fb26ec6c(

Password

8pbrEyfTWRka

Project ID

qwiklabs-gcp-00-0b0924b!

## Upload CSV files to tables

Return to the Cloud SQL console. You will now upload the `start_station_name` and `end_station_name` CSV files into your newly created london1 and london2 tables.

1. In your Cloud SQL instance page, click **IMPORT**.
2. In the Cloud Storage file field, click **Browse**, and then click the arrow next to your bucket name, and then click `start_station_data.csv`. Click **Select**.
3. Select **CSV** as File format.
4. Select the `bike` database and type in `london1` as your table.
5. Click **Import**.

Do the same for the other CSV file.

1. In your Cloud SQL instance page, click **IMPORT**.

100/100

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Lab setup instructions and requirements

End Lab 00:44:17

**Caution:** When you are in the console, do not deviate from the lab instructions. Doing so may cause your account to be blocked. Learn more.

Open Google Cloud console

Username
student-00-d949fb26ec6c(

Password
8pbrEyfTWRka

Project ID
qwiklabs-gcp-00-0b0924b!

bucket name, and then click `start_station_data.csv`. Click **Select**.

3. Select **CSV** as File format.
4. Select the `bike` database and type in `london1` as your table.
5. Click **Import**.

Do the same for the other CSV file.

1. In your Cloud SQL instance page, click **IMPORT**.
2. In the Cloud Storage file field, click **Browse**, and then click the arrow next to your bucket name, and then click `end_station_data.csv` Click **Select**.
3. Select **CSV** as File format.
4. Select the `bike` database and type in `london2` as your table.
5. Click **Import**.

You should now have both CSV files uploaded to tables in the `bike` database.

100/100

my-demo – Overview – Cloud SQL | gcloud CLI Remote Login

https://console.cloud.google.com/sql/instances/my-demo/overview?cloudshell=true&project=qwiklabs-gcp-00-0b0924b5d37c

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Google Cloud | qwiklabs-gcp-00-0b0924b5d37c | cloud sql | Search

Cloud SQL

Overview | Edit | Import | Export | Restart | Stop | Delete | Clone | Failover

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- Query insights
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- Databases
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- Replicas
- Operations

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All instances > my-demo

# my-demo

MySQL 8.0

## Learn the basics of Cloud SQL

0 of 3 completed

1. Import data
2. Query and explore data
3. Connect source application

With your Cloud SQL instance now running, the next step is to import data. You can import CSV and SQL files from your local file system or from a GCS bucket.

If you don't have existing data, you can create a database, then create a table in Cloud SQL Studio in the next step.

Import data | Mark as complete

1 hour | 6 hours | 1 day | 7 days | 30 days | Custom

Chart

CPU utilization


CLOUD SHELL

Terminal (qwiklabs-gcp-00-0b0924b5d37c)

Open Editor

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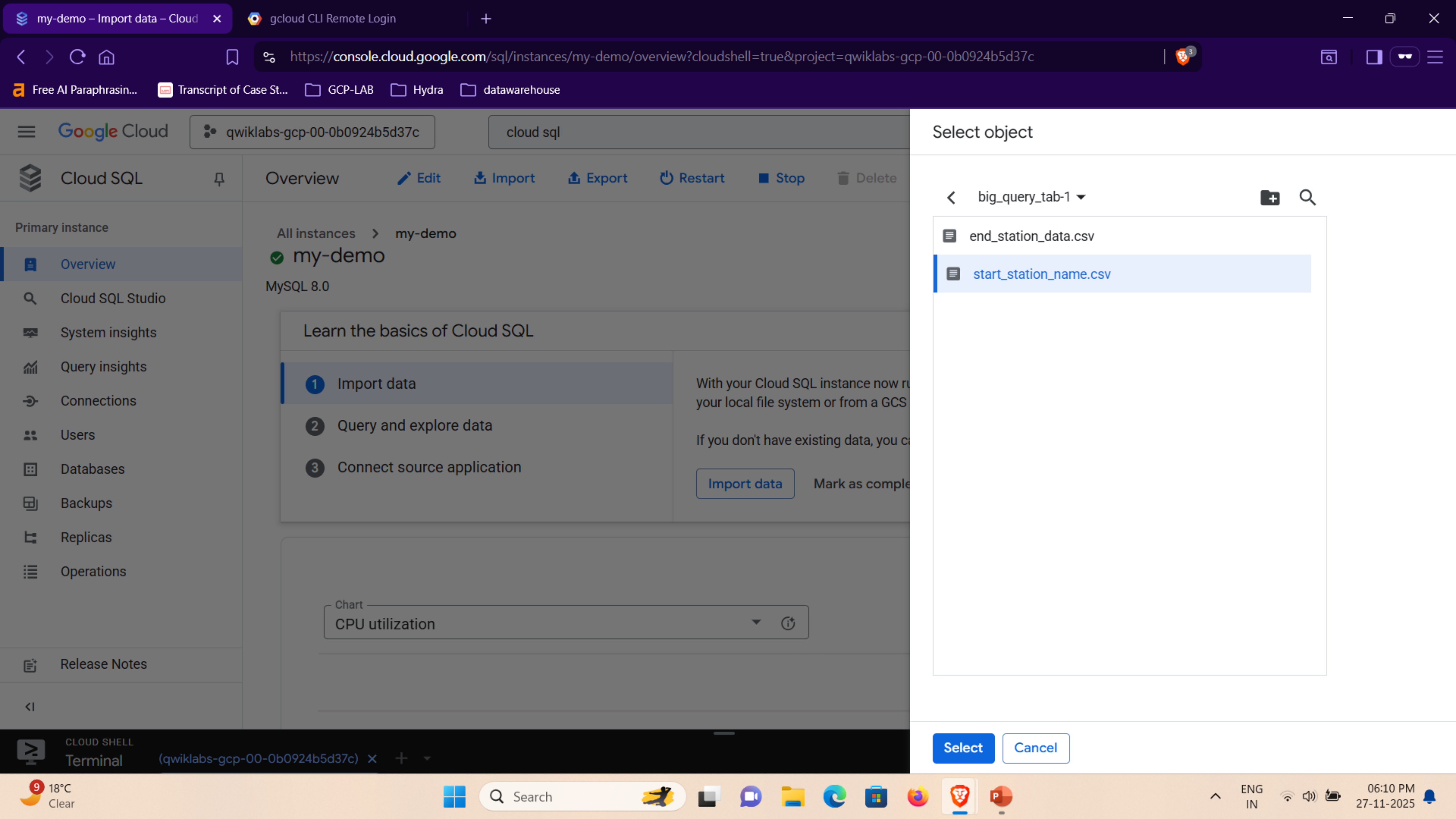
my-demo – Import data – Cloud
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your local file system or from a GCS
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Import data
Mark as comple
Chart
CPU utilization
Select object
big_query_tab-1
end_station_data.csv
start_station_name.csv
Select
Cancel
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(qwiklabs-gcp-00-0b0924b5d37c)
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27-11-2025

my-demo – Cloud SQL – qwiklab

gcloud CLI Remote Login

https://console.cloud.google.com/sql/instances/my-demo/overview?cloudshell=true&project=qwiklabs-gcp-00-0b0924b5d37c

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Google Cloud | qwiklabs-gcp-00-0b0924b5d37c | cloud sql | Search

Cloud SQL

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All instances > my-demo

# my-demo

MySQL 8.0

## Learn the basics of Cloud SQL

0 of 3 completed

1. Import data
2. Query and explore data
3. Connect source application

With your Cloud SQL instance now running, the next step is to import data. You can import CSV and SQL files from your local file system or from a GCS bucket.

If you don't have existing data, you can create a database, then create a table in Cloud SQL Studio in the

Import data | Mar

1 hour

### Uploads and qwiklabs-gcp-00-0b09... operations

Imported from start_station_name.csv to my-demo — 6:11:11 PM GMT+5

Created my-demo. — 5:58:59 PM GMT+5

Uploading 2 files

- end_station_data.csv — Complete
- start_station_name.csv — Complete

CLOUD SHELL

Terminal (qwiklabs-gcp-00-0b0924b5d37c)

Open Editor

18°C Clear | Search | ENG IN | 06:11 PM 27-11-2025

my-demo – Cloud SQL – qwiklab | gcloud CLI Remote Login

https://console.cloud.google.com/sql/instances/my-demo/overview?cloudshell=true&project=qwiklabs-gcp-00-0b0924b5d37c

Free AI Paraphrasin... | Transcript of Case St... | GCP-LAB | Hydra | datawarehouse

Google Cloud | qwiklabs-gcp-00-0b0924b5d37c | cloud sql | Search

Cloud SQL

Overview | Edit | Import | Export | Restart | Stop | Delete | Clone | Failover

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All instances > my-demo

# my-demo

MySQL 8.0

## Learn the basics of Cloud SQL

0 of 3 completed

1. Import data
2. Query and explore data
3. Connect source application

With your Cloud SQL instance now running, the next step is to import data. You can import CSV and SQL files from your local file system or from a GCS bucket.

If you don't have existing data, you can create a database, then create a table in Cloud SQL Studio in the

Import data | Mar

1 hour

**Uploads and qwiklabs-gcp-00-0b09... operations**

| | |
|---|---|
| Imported from end_station_data.csv to my-demo | 6:12:24 PM GMT+5 |
| Imported from start_station_name.csv to my-demo | 6:11:11 PM GMT+5 |
| Created my-demo. | 5:58:59 PM GMT+5 |

Uploading 2 files

CLOUD SHELL
Terminal (qwiklabs-gcp-00-0b0924b5d37c) | Open Editor

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Introduction to SQL for BigQuery and Cloud SQL

Lab setup instructions and requirements

End Lab 00:38:55

**Caution:** When you are in the console, do not deviate from the lab instructions. Doing so may cause your account to be blocked. Learn more.

Open Google Cloud console

**Username**
student-00-d949fb26ec6c(

**Password**
8pbrEyfTWRka

**Project ID**
qwiklabs-gcp-00-0b0924b!

You should now have both CSV files uploaded to tables in the `bike` database.

1. Return to your Cloud Shell session and run the following command at the MySQL server prompt to inspect the contents of london1:

```
SELECT * FROM london1;
```

You should receive 955 lines of output, one for each unique station name.

2. Run the following command to make sure that london2 has been populated:

```
SELECT * FROM london2;
```

You should receive 959 lines of output, one more each unique station name.

## DELETE keyword

```
mysql> SELECT * FROM london1;
+--------------------------------------------------------+--------+
| start_station_name                                     | num    |
+--------------------------------------------------------+--------+
| start_station_name                                     |      0 |
| Hyde Park Corner, Hyde Park                            | 671688 |
| Belgrove Street , King's Cross                         | 593065 |
| Waterloo Station 3, Waterloo                           | 527112 |
| Albert Gate, Hyde Park                                 | 461108 |
| Black Lion Gate, Kensington Gardens                    | 459716 |
| Waterloo Station 1, Waterloo                           | 419334 |
| Wellington Arch, Hyde Park                             | 392889 |
| Hop Exchange, The Borough                              | 385926 |
| Wormwood Street, Liverpool Street                      | 354663 |
| Triangle Car Park, Hyde Park                           | 353033 |
| Duke Street Hill, London Bridge                        | 334976 |
| Brushfield Street, Liverpool Street                    | 324478 |
| Bethnal Green Road, Shoreditch                         | 319191 |
| Aquatic Centre, Queen Elizabeth Olympic Park           | 306440 |
| Craven Street, Strand                                  | 293522 |
| Storey's Gate, Westminster                             | 291048 |
| Park Lane , Hyde Park                                  | 282810 |
| Serpentine Car Park, Hyde Park                         | 280172 |
| Palace Gate, Kensington Gardens                        | 276525 |
| Tooley Street, Bermondsey                              | 269700 |
| Soho Square , Soho                                     | 269552 |
| Cheapside, Bank                                        | 268747 |
| Tower Gardens , Tower                                  | 266016 |
| Finsbury Circus, Liverpool Street                      | 263926 |
| Queen Street 1, Bank                                   | 261950 |
| Whitehall Place, Strand                                | 258146 |
| Exhibition Road, Knightsbridge                         | 255008 |
| Green Park Station, Mayfair                            | 253217 |
| Crosswall, Tower                                       | 251907 |
| Waterloo Station 2, Waterloo                           | 250920 |
| Shoreditch High Street, Shoreditch                     | 246615 |
| Little Argyll Street, West End                         | 242981 |
| Vauxhall Cross, Vauxhall                               | 242599 |
```

my-demo – Cloud SQL – qwiklab
gcloud CLI Remote Login

https://console.cloud.google.com/sql/instances/my-demo/overview?cloudshell=true&project=qwiklabs-gcp-00-0b0924b5d37c

Free AI Paraphrasin... Transcript of Case St... GCP-LAB Hydra datawarehouse

```
| St. Mary and St. Michael Church, Stepney                |    820 |
| Central House, Whitechapel                              |    817 |
| Halford Road, Fulham                                    |    785 |
| Westfield Eastern Access Road, Shepherd's Bush          |    778 |
| Allington street, Westminster, SW1E 5EB                 |    704 |
| Spanish Road, Wandsworth                                |    666 |
| Stewart's Road, Nine Elms                               |    663 |
| Harris Academy, Bermondsey                              |    656 |
| Ingrave Street, Battersea                               |    635 |
| Lansdown Drive, Hackney Central                         |    630 |
| South Bermondsey Station, Bermondsey                    |    608 |
| Chicheley Street, South Bank                            |    555 |
| Queensway, Paddington                                   |    522 |
| The Blue, Bermondsey                                    |    483 |
| Limburg Road, Clapham Common                            |    466 |
| St John's Park, Cubitt Town                             |    449 |
| Mechanical Workshop Penton                              |    392 |
| Coborn Street, Mile End                                 |    340 |
| Exhibition Road, South Kensington                       |    270 |
| Import Dock                                             |    142 |
| Crimscott Street, Bermondsey                            |    139 |
| Pop Up Dock 1                                           |    126 |
| Wansey Street, Walworth                                 |     97 |
| Cartier Circle, Canary Wharf                            |     75 |
| Pop Up Dock 2                                           |     70 |
| Contact Centre, Southbury House                         |     60 |
| Allington street, Off Victoria Street, Westminster      |     55 |
| Monier Road, Newham                                     |     34 |
| Monier Road                                             |     11 |
| LSP2                                                    |      7 |
| Blackfriars road, Southwark                             |      5 |
| One London                                              |      4 |
| LSP1                                                    |      2 |
| York Way, Camden                                        |      1 |
| PENTON STREET COMMS TEST TERMINAL _ CONTACT MATT McNULTY |      1 |
+---------------------------------------------------------+--------+
955 rows in set (0.41 sec)

mysql>
```

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https://console.cloud.google.com/sql/instances/my-demo/overview?cloudshell=true&project=qwiklabs-gcp-00-0b0924b5d37c

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```
mysql> SELECT * FROM london2;
+----------------------------------------------------------+--------+
| end_station_name                                         | num    |
+----------------------------------------------------------+--------+
| end_station_name                                         |      0 |
| Hyde Park Corner, Hyde Park                              | 671680 |
| Belgrove Street , King's Cross                           | 586568 |
| Hop Exchange, The Borough                                | 519033 |
| Waterloo Station 3, Waterloo                             | 508421 |
| Albert Gate, Hyde Park                                   | 464713 |
| Black Lion Gate, Kensington Gardens                      | 452412 |
| Waterloo Station 1, Waterloo                             | 403861 |
| Wellington Arch, Hyde Park                               | 388354 |
| Brushfield Street, Liverpool Street                      | 380069 |
| Wormwood Street, Liverpool Street                        | 372718 |
| Triangle Car Park, Hyde Park                             | 349623 |
| Duke Street Hill, London Bridge                          | 349479 |
| Bethnal Green Road, Shoreditch                           | 322209 |
| Storey's Gate, Westminster                               | 318794 |
| Soho Square , Soho                                       | 314085 |
| Aquatic Centre, Queen Elizabeth Olympic Park             | 309576 |
| Holborn Circus, Holborn                                  | 308103 |
| Craven Street, Strand                                    | 307151 |
| Tooley Street, Bermondsey                                | 300325 |
| St. James's Square, St. James's                          | 294442 |
| Finsbury Circus, Liverpool Street                        | 291985 |
| Whitehall Place, Strand                                  | 287426 |
| Cheapside, Bank                                          | 284815 |
| Tower Gardens , Tower                                    | 284369 |
| Park Lane , Hyde Park                                    | 282455 |
| Serpentine Car Park, Hyde Park                           | 281158 |
| Exhibition Road, Knightsbridge                           | 279276 |
| Newgate Street , St. Paul's                              | 278570 |
| Queen Street 1, Bank                                     | 277824 |
| Palace Gate, Kensington Gardens                          | 277231 |
| Green Park Station, Mayfair                              | 270166 |
| Crosswall, Tower                                         | 267956 |
| Little Argyll Street, West End                           | 264761 |
```

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```
| Harrington Square, Camden Town                            |    740 |
| Mechanical Workshop Penton                                |    724 |
| Harris Academy, Bermondsey                                |    666 |
| South Bermondsey Station, Bermondsey                      |    665 |
| Lansdown Drive, Hackney Central                           |    649 |
| Ingrave Street, Battersea                                 |    639 |
| Lansdowne Walk, Notting Hill                              |    602 |
| Wansey Street, Walworth                                   |    594 |
| Chicheley Street, South Bank                              |    589 |
| Stewart's Road, Nine Elms                                 |    572 |
| Spanish Road, Wandsworth                                  |    537 |
| Ladbroke Grove Central, Notting Hill                      |    526 |
| Queensway, Paddington                                     |    510 |
| The Blue, Bermondsey                                      |    508 |
| Limburg Road, Clapham Common                              |    481 |
| St John's Park, Cubitt Town                               |    393 |
| Coborn Street, Mile End                                   |    315 |
| Exhibition Road, South Kensington                         |    260 |
| Import Dock                                               |    146 |
| Crimscott Street, Bermondsey                              |    133 |
| Pop Up Dock 1                                             |    110 |
| Contact Centre, Southbury House                           |     60 |
| Cartier Circle, Canary Wharf                              |     58 |
| Pop Up Dock 2                                             |     52 |
| Allington street, Off Victoria Street, Westminster        |     51 |
| Monier Road, Newham                                       |     36 |
| Monier Road                                               |     14 |
| PENTON STREET COMMS TEST TERMINAL _ CONTACT MATT McNULTY |     11 |
| LSP1                                                      |      7 |
| LSP2                                                      |      5 |
| Blackfriars road, Southwark                               |      5 |
| One London                                                |      4 |
| Electrical Workshop PS                                    |      2 |
| Canada Water Station                                      |      1 |
| York Way, Camden                                          |      1 |
+-----------------------------------------------------------+--------+
959 rows in set (0.22 sec)

mysql>
```

Lab setup instructions and requirements

End Lab 00:36:24

**Caution:** When you are in the console, do not deviate from the lab instructions. Doing so may cause your account to be blocked. Learn more.

Open Google Cloud console

**Username**

student-00-d949fb26ec6c(

**Password**

8pbrEyfTWRka

**Project ID**

qwiklabs-gcp-00-0b0924b!

You should receive 959 lines of output, one more each unique station name.

## DELETE keyword

Here are a couple more SQL keywords that help us with data management. The first is the `DELETE` keyword.

- Run the following commands in your MySQL session to delete the first row of the london1 and london2:

Copied!

```
DELETE FROM london1 WHERE num=0;
DELETE FROM london2 WHERE num=0;
```

You should receive the following output after running both commands:

```
Query OK, 1 row affected (0.04 sec)
```

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```
| Lansdowne Walk, Notting Hill                            |    602 |
| Wansey Street, Walworth                                 |    594 |
| Chicheley Street, South Bank                            |    589 |
| Stewart's Road, Nine Elms                               |    572 |
| Spanish Road, Wandsworth                                |    537 |
| Ladbroke Grove Central, Notting Hill                    |    526 |
| Queensway, Paddington                                   |    510 |
| The Blue, Bermondsey                                    |    508 |
| Limburg Road, Clapham Common                            |    481 |
| St John's Park, Cubitt Town                             |    393 |
| Coborn Street, Mile End                                 |    315 |
| Exhibition Road, South Kensington                       |    260 |
| Import Dock                                             |    146 |
| Crimscott Street, Bermondsey                            |    133 |
| Pop Up Dock 1                                           |    110 |
| Contact Centre, Southbury House                         |     60 |
| Cartier Circle, Canary Wharf                            |     58 |
| Pop Up Dock 2                                           |     52 |
| Allington street, Off Victoria Street, Westminster      |     51 |
| Monier Road, Newham                                     |     36 |
| Monier Road                                             |     14 |
| PENTON STREET COMMS TEST TERMINAL _ CONTACT MATT McNULTY |     11 |
| LSP1                                                    |      7 |
| LSP2                                                    |      5 |
| Blackfriars road, Southwark                             |      5 |
| One London                                              |      4 |
| Electrical Workshop PS                                  |      2 |
| Canada Water Station                                    |      1 |
| York Way, Camden                                        |      1 |
+---------------------------------------------------------+--------+
959 rows in set (0.22 sec)

mysql> DELETE FROM london1 WHERE num=0;
Query OK, 1 row affected (0.22 sec)

mysql> DELETE FROM london2 WHERE num=0;
Query OK, 1 row affected (0.21 sec)

mysql>
```

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Lab setup instructions and requirements

End Lab 00:35:58

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Open Google Cloud console

**Username**
student-00-d949fb26ec6c(

**Password**
8pbrEyfTWRka

**Project ID**
qwiklabs-gcp-00-0b0924b!

## INSERT INTO keyword

You can also insert values into tables with the `INSERT INTO` keyword.

- Run the following command to insert a new row into london1, which sets `start_station_name` to "test destination" and `num` to "1":

```
INSERT INTO london1 (start_station_name, num) VALUES ("test destination", 1);
```

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The `INSERT INTO` keyword requires a table (london1) and will create a new row with columns specified by the terms in the first parenthesis (in this case "start_station_name" and "num"). Whatever comes after the "VALUES" clause will be inserted as values in the new row.

You should receive the following output:

```
Query OK, 1 row affected (0.05 sec)
```

```
| Stewart's Road, Nine Elms                             |    572 |
| Spanish Road, Wandsworth                              |    537 |
| Ladbroke Grove Central, Notting Hill                  |    526 |
| Queensway, Paddington                                 |    510 |
| The Blue, Bermondsey                                  |    508 |
| Limburg Road, Clapham Common                          |    481 |
| St John's Park, Cubitt Town                           |    393 |
| Coborn Street, Mile End                               |    315 |
| Exhibition Road, South Kensington                     |    260 |
| Import Dock                                           |    146 |
| Crimscott Street, Bermondsey                          |    133 |
| Pop Up Dock 1                                         |    110 |
| Contact Centre, Southbury House                       |     60 |
| Cartier Circle, Canary Wharf                          |     58 |
| Pop Up Dock 2                                         |     52 |
| Allington street, Off Victoria Street, Westminster    |     51 |
| Monier Road, Newham                                   |     36 |
| Monier Road                                           |     14 |
| PENTON STREET COMMS TEST TERMINAL _ CONTACT MATT McNULTY |     11 |
| LSP1                                                  |      7 |
| LSP2                                                  |      5 |
| Blackfriars road, Southwark                           |      5 |
| One London                                            |      4 |
| Electrical Workshop PS                                |      2 |
| Canada Water Station                                  |      1 |
| York Way, Camden                                      |      1 |
+-------------------------------------------------------+--------+
959 rows in set (0.22 sec)

mysql> DELETE FROM london1 WHERE num=0;
Query OK, 1 row affected (0.22 sec)

mysql> DELETE FROM london2 WHERE num=0;
Query OK, 1 row affected (0.21 sec)

mysql> INSERT INTO london1 (start_station_name, num) VALUES ("test destination", 1);
Query OK, 1 row affected (0.22 sec)

mysql>
```

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Introduction to SQL for BigQuery and Cloud SQL

Lab setup instructions and requirements

End Lab 00:35:30

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Open Google Cloud console

Username
student-00-d949fb26ec6c(

Password
8pbrEyfTWRka

Project ID
qwiklabs-gcp-00-0b0924b!

## UNION keyword

The last SQL keyword that you'll learn about is `UNION`. This keyword combines the output of two or more `SELECT` queries into a result-set. You use `UNION` to combine subsets of the "london1" and "london2" tables.

The following chained query pulls specific data from both tables and combines them with the `UNION` operator.

- Run the following command at the MySQL server prompt:

```
SELECT start_station_name AS top_stations, num FROM london1
WHERE num>100000
UNION
SELECT end_station_name, num FROM london2 WHERE num>100000
ORDER BY top_stations DESC;
```

The first `SELECT` query selects the two columns from the "london1" table and creates an alias for "start_station_name", which gets set to "top_stations". It uses the `WHERE`

```
mysql> SELECT start_station_name AS top_stations, num FROM london1 WHERE num>100000
    -> UNION
    -> SELECT end_station_name, num FROM london2 WHERE num>100000
    -> ORDER BY top_stations DESC;
+----------------------------------------------+--------+
| top_stations                                 | num    |
+----------------------------------------------+--------+
| York Hall, Bethnal Green                     | 119347 |
| York Hall, Bethnal Green                     | 122406 |
| Wright's Lane, Kensington                    | 151734 |
| Wright's Lane, Kensington                    | 161159 |
| Wren Street, Holborn                         | 134375 |
| Wren Street, Holborn                         | 123859 |
| Wormwood Street, Liverpool Street            | 354663 |
| Wormwood Street, Liverpool Street            | 372718 |
| Woodstock Grove, Shepherd's Bush             | 131922 |
| Woodstock Grove, Shepherd's Bush             | 134718 |
| Winsland Street, Paddington                  | 118176 |
| Winsland Street, Paddington                  | 120951 |
| Windsor Terrace, Hoxton                      | 105464 |
| William IV Street, Strand                    | 210375 |
| William IV Street, Strand                    | 257283 |
| Whitehall Place, Strand                      | 287426 |
| Whitehall Place, Strand                      | 258146 |
| Westminster Pier, Westminster                | 146382 |
| Westminster Pier, Westminster                | 148627 |
| Westferry DLR, Limehouse                     | 134608 |
| Westferry DLR, Limehouse                     | 147288 |
| Westferry Circus, Canary Wharf               | 106211 |
| Westferry Circus, Canary Wharf               | 110809 |
| Westbridge Road, Battersea                   | 153756 |
| Westbridge Road, Battersea                   | 151111 |
| Westbourne Grove, Bayswater                  | 101600 |
| West Smithfield Rotunda, Farringdon          | 162833 |
| West Smithfield Rotunda, Farringdon          | 156530 |
| Wenlock Road , Hoxton                        | 128828 |
| Wenlock Road , Hoxton                        | 104741 |
| Wells Street, Fitzrovia                      | 139262 |
```

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```
| Bath Street, St. Luke's                       | 163819 |
| Bath Street, St. Luke's                       | 161274 |
| Barons Court Station, West Kensington         | 113920 |
| Barons Court Station, West Kensington         | 108586 |
| Bankside Mix, Bankside                        | 201623 |
| Bankside Mix, Bankside                        | 174110 |
| Bank of England Museum, Bank                  | 173370 |
| Bank of England Museum, Bank                  | 186018 |
| Baldwin Street, St. Luke's                    | 110713 |
| Baldwin Street, St. Luke's                    | 111383 |
| Ashley Place, Victoria                        | 175144 |
| Ashley Place, Victoria                        | 183860 |
| Aquatic Centre, Queen Elizabeth Olympic Park  | 309576 |
| Aquatic Centre, Queen Elizabeth Olympic Park  | 306440 |
| Ansell House, Stepney                         | 103641 |
| Ansell House, Stepney                         | 110740 |
| Allington Street, Victoria                    | 128336 |
| Allington Street, Victoria                    | 129176 |
| All Saints Church, Portobello                 | 139718 |
| All Saints Church, Portobello                 | 137984 |
| Aldersgate Street, Barbican                   | 190311 |
| Aldersgate Street, Barbican                   | 183256 |
| Alderney Street, Pimlico                      | 128339 |
| Alderney Street, Pimlico                      | 120167 |
| Albert Gate, Hyde Park                        | 464713 |
| Albert Gate, Hyde Park                        | 461108 |
| Albert Bridge Road, Battersea Park            | 131622 |
| Albert Bridge Road, Battersea Park            | 127936 |
| Albany Street, The Regent's Park              | 101765 |
| Ada Street, Hackney Central                   | 106514 |
| Ada Street, Hackney Central                   | 111076 |
| Abingdon Green, Westminster                   | 159305 |
| Abingdon Green, Westminster                   | 134157 |
| Abbey Orchard Street, Westminster             | 121574 |
| Abbey Orchard Street, Westminster             | 145521 |
+-----------------------------------------------+--------+
629 rows in set (0.21 sec)

mysql>
```

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Search
ENG
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